

Individual Project: Trucking Dispatching System - Price Transparency Tool

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Abstract— we will cover the initial discovery for the Trucking Dispatching System, specifically the cost transparency feature. In this document we will cover the motivation behind the tool and progress in discovery.

1 TRUCKING DISPATCH SYSTEM INTRODUCTION

The logistics industry generates \$8 trillion to \$12 trillion annually and contributes up to 10 percent to the U.S. gross domestic product (GDP). Shippers ship freight throughout the country using all forms of transportation such as trains, trucks, boats, and airplanes. The shipment industry is massive and supports all forms of business from small business up to companies that are listed in the S&P 500. Specifically, the trucking logistics market generates roughly \$4.1 billion. Working in the logistic industry in the corporate space, I identified areas of opportunity for technology to be adopted that is more simplistic and informative. A few of my relatives and close friends established a trucking business that operated throughout the U.S. but specifically in Texas and throughout the midwest. I have been able to observe the logistics industry in many different forms such as riding with drivers to deliver freight, work in the shipment facility to unload packages, and currently working in the corporate space working on data transformation.

Although the logistic industry is evolving, there are areas of opportunity that technology could help close gaps. One specific area is informative freight pricing for the shipper and truckers that use third party dispatchers to complete freight movement. Currently, some small businesses use dispatching companies to contract truckers to complete freight shipment. Price and cost fluctuation is common with this industry which impacts the cost to ship freight which would impact the shippers and profit margins for truckers. There are many variables that

impact trucking profit margins such as operating costs such as gas prices, insurance pricing, and trucking maintenance. For shippers prices can vary based on the type of freight, weight, distance, and timing. Based on my observation, I would like to investigate the pricing transparency for shippers and truckers.

For the shipping price validation, I would like to build a product that provides visibility to both shippers and truckers related to pricing calculation based on the type of freight and distance. The feature will be informative and showcase how much shippers should pay for shipping freight and informative to truckers related to revenue for accepting freight. The tool will be peer to peer which would remove the need for dispatching and increase cost savings for shippers and additional profits for truckers. The goal of the tool is for it to be easy to adopt, cost effective for both users, and informative.

2 NEEDFINDING/DISCOVERY PROCESS

The overall objective is to gain perspective on how the different individuals operate and navigate the logistics industry. The two initial target users for interviews one will be truckers that have operated their own trucking business and had direct interaction with dispatching. The second are industry professionals that have extensive experience building logistics infrastructures, operated closely with contractors or dispatch, and technical individuals that develop software that is used within the industry today. The objective is to understand the different individuals skillset and gain perspective on the operations that occur when shipping, managing, and delivering freight.

2.1 User interview Questions for Truckers / Operators

The questions below will be the initial questions to guide the conversation regarding the operations for logistics. The goal is to gain a perspective on freight pricing, personal experiences when working with a dispatching company, and how economical conditions impact the cost to deliver freight. The questions below will be iterated on for more open-ended discussions.

2.2 User Interview Questions for Experience Logistic Employees

The objective is to gain perspective from individuals that work for logistic companies. The cost to ship products and goods is one of the foundations of revenue for logistic companies. The goal is to understand if common individuals that work in the logistic space understand the cost breakdown of shipping goods. Is there transparency regarding industry shifts and how economical impacts change operations. The true goal is do individuals that build logistic systems, interact with consumers, or drive strategic initiatives have a general understanding on how cost is calculated to ship products and goods from one location to another. The interview questions are conversation starters to gain a better perspective of the individual.

2.2.1 User Interviews Recipients

I have identified hour truckers that have owned their own trucking business or are current employees of a trucking business. For the experienced logistic employees, I have six different individuals that will participate in user interviews. All individuals have agreed to provide support throughout this process. The recipients would range from individuals that have an E-commerce business, operated for a carrier as an employee, owns a carrier business, and traditional shippers.

2.3 Heuristic Evaluations

There are several heuristic evaluations that could be identified for the Trucking Dispatch - Pricing Transparency tool. The three I will focus on will be visibility of system status, recognition rather than recall, and error prevention. Currently there is a lack of visibility in how prices are calculated when shipping freight or an understanding of price fluctuation. Users (drivers and shippers) are only presented with the final pricing number related to freight cost. For truckers the heuristic evaluation would be error prevention since inexperienced drivers would not have knowledge of fuel calculation when factoring freight cost. Dispatchers would have a set price and drivers would not have an idea on how to gauge if the freight would be profitable. Drivers run the risk of not properly evaluating routes

that would not be profitable. The recognition rather than recall heuristic evaluation for users is that individuals would need to cross check rates using multiple tools, data points, and trusting sources. The cognitive overload would impact users' decisions and present missed opportunities.

2.4 Needfinding Results

There are two opportunities for logistics pricing transparency based on the user interviews. One is for shippers, users really do not have an idea on how shipping cost is calculated, even the experienced logistic employees do not fully understand the true cost to ship a package from point A to point B. Price factoring is not a huge factor for everyday consumers. The type of package would determine how a user would select a carrier to ship a package. Users that do not ship often, really do not care about the final number, but reliability would outweigh the cost to ship a package.

Shippers that operate a small business do care about the cost to ship packages and would like to compare carriers based on the type of packages that are being shipped and arrival time. Everyday shippers would like to see more details outside of pricing transparency, users would like to view a carrier's reliability compared to other carriers. A tool that does show shipping price transparency is a tool that would be used.

Drivers and small logistic operators would need more transparent data for freight cost. Drivers would like to have trusted indicators that would be a part of the calculation to factor in for freight cost transparency. Currently, drivers and operators are only presented with the final number for freight transportation. Drivers that are not large carriers or shippers are not able to negotiate discounts since the volume determines the price. However, there is an opportunity to provide transparency for fairness.

2.4.1 Needfinding Results - User interviews

I was able to get a total of 7 individuals to participate in the user interviews so far with a range of experience: logistic corporate employees, logistic business owners, and drivers. The results of the interviews provide me insight on the difference

between your everyday shippers which are typically business owners and your casual shipper. The outcome is a user friendly platform that is easy to use and provides users an inviting feel once while using the platform. The biggest hurdle would be user attraction and providing a quick and effective solution that provides users the access of data to make effective decisions while comparing shippers.

Since the common users would use this platform once or twice per year then it would be important to provide quick and valuable data for consumers. The average shipper would want a platform that allows them to make decisions that are nonbias or provides information outside of their immediate needs. The carriers have expressed interesting information that allows them to make decisions on the type of freight they could haul. One issue is the deadmiles which are miles driven that do not generate profits. If a carrier ships a location that is eight hours to the delivery destination, then it would be helpful to understand how many empty miles be factored in the completion of the delivery to help with negotiation tactics.

While riding along with one of the users earlier last year, I also recalled an opportunity that would help carriers and truck owners and that is the generative AI component. The AI component would work as an assistance to help maximize profits to small businesses that would be affordable. Factoring in the operating cost such as insurance and fuel to help determine the profit margins. The information collected will influence the brainstorming process.

3 BRAINSTORMING

Using the information gathered from the users interviews, I will be able to investigate freight calculators for current functionality and apply the areas of opportunities that were identified in the user interviews. I also researched similar calculators that provide shippers data for estimated cost to ship products. The freight negotiation feature is a concept that would be biasless and truly focus on the carriers. Based on the research, most of the apps are company first, carrier second which creates a disadvantage when accepting freights to be shipped for a company. The goal is to provide carriers more than enough information to

evaluate freight and make effective business decisions before accepting the job. The examples below are the company focused freight calculators.

The ArcBest Freight Calculator:

New Quote

PICKUP LOCATION

Type Business

Country US Postal Code

Pickup Services

DELIVERY LOCATION

Type Business

Country US Postal Code

Delivery Services

PICKUP DATE MM/DD/YYYY

Pickup Date 10/31/2025

QUOTE REQUESTER

Work Email (optional)

HANDLING UNIT 1

Type Pallet Qty. 1

Stack? ☒

L W H

WT.

Commodity Description

Class

Haz.

☐ Inches

☐ Pounds

H/U SUBTOTAL: -- PCF 0 lbs.

Compare quotes for LTL, Time Critical, Volume and FTL with a single click!

Get Instant Rates

New Quote

HANDLING UNIT 1

Type Pallet Qty. 1

Stack? ☒

L W H

WT.

Commodity Description

Class

Haz.

☐ Inches

☐ Pounds

H/U SUBTOTAL: -- PCF 0 lbs.

Pallet

Bag

Bale

Barrel

Basket

Box

Bucket

Compare quotes for LTL, Time Critical, Volume and FTL with a single click!

Get Instant Rates

New Quote

Total: 1 H/U 0 cu. ft. 0 PCF 0 lbs.

☐ Single Shipment

Select when your single shipment will be picked up at one time and place, unaccompanied by any other shipments from the same pickup site.

☐ Premium Cargo OR Excess Liability Coverage

Select when additional cargo liability is needed, and enter the declared value of your shipment.
[Learn more](#)

☐ Sort & Segregate

Select when additional or special handling of the shipment by ABF personnel is requested or ABF is requested to sort or segregate the contents of the shipment by marks, brands, sizes, flavors or other distinguishing characteristics.

☐ Customs or In-Bond Freight

Select when your shipment is moving under United States Customs Bond for clearance at a point in the United States, or requiring delivery under U.S. Customs supervision. You may review Item 480 ABF 111 for more details.
[Learn more](#)

☐ Freeze Protection

Select when your shipment will require protection from freezing temperatures (service availability subject to certain restrictions).
[Learn more](#)

Compare quotes for LTL, Time Critical, Volume and FTL with a single click!

Get Instant Rates

6

FedEx Freight Calculator:

Calculate FedEx shipping rates

From*

Milwaukee, Wisconsin, 53202, United States

To*

Seattle, Washington, 98154, United States

53202 and 98154 are the closest matches we could find. If they're wrong, you can change them in the fields above.

☐

I'm shipping to a residential address

Tell us more about your shipment

ASK FEDEX

Tell us more about your shipment

Packaging*

Your Packaging

☐

Purchase a higher limit of liability from FedEx

Packages*

1

Package weight*

lb

Dimensions L x W x H

x

x

in

+ ADD ANOTHER PACKAGE

Ship date*

Friday, October 31, 2025

FedEx® Retail Rates

FedEx One Rate®

Here are the FedEx® Retail Rates and delivery dates for shipments sent Friday, October 31, 2025

☒

I want this shipment picked up

☒

Schedule a pickup

☐

Use an already scheduled pickup

☐

View signature options

☐

I'm shipping non-standard package(s)

7

Amounts are shown in USD

ARRIVES ON Sat, Nov 1	DELIVERED BY 9:30 AM FedEx First Overnight®	\$382.80	▼
	DELIVERED BY 12:00 PM FedEx Priority Overnight®	\$345.91	▼
	DELIVERED BY 5:00 PM FedEx Standard Overnight®	\$337.28	▼
ARRIVES ON Mon, Nov 3	DELIVERED BY 8:00 AM FedEx First Overnight®	\$363.76	▼

3.1 Brainstorming Plan

After several brainstorming sessions the individual brainstorm by partnering with ChatGPT and chainstorming from information gathered from the user interviews. There were a few features to focus on for the consumers that used shipping as a service and truckers.

The generated ideas:

1. Transparency and Trust
2. Education and Awareness
3. Negotiations
4. Convenience
5. Community Reviews

The area that I will focus on is the pricing clarity, freight fairness, and usability. The concept is to have it optional for users to create an account or sign-in in order to use the common features. One of the key findings is that the rate calculator specifically is tailored to the company that owns the website. The solution I am proposing is biasless and is open to all companies for better selection that shippers would be able to view.

3.2 Brainstorming Results

3.2.1 Freight & Shipping Pricing Transparency Page

The Freight & Shipping Pricing Transparency Page would be for both the customers that ship packages, and the truckers ship freight. The page would provide a cost breakdown on how much it would cost to ship a package to a location, or what would be the fair rate for freight that would be shipped by the carrier. The common components of weight, distance, speed, and potential additional changes such as toll cost, and gas price changes depending on the state would provide both users a detailed price breakdown for the price.

The freight & Shipping page would directly address one of the key issues with the lack of price transparency and limited trust in freight opportunity cost for carriers. It provides visibility for both users by providing carriers with pay fairness data, and shippers with rate justification. The design goals would be to simplify the freight variables into visuals. Allow users to simulate “what if” scenarios, and present a rate fairness index for carriers.

An opportunity that the freight & Shippers page would provide is the ability to find unlikely carrier companies to handle the shipper needs. Consumers would have access to the ideal carriers based on their needs rather than the most popular or the consumer that pays the most to be atop the algorithm. The market trends would provide users the ability to view if the calculated price is on par or above the market rates to ship packages.

3.2.2 Freight Negotiation Page

The Freight Negotiation Page would be a digital assistance page that would help carriers create counter offers by analyzing the route data, package information, market trends, and ideal profit margins. The negotiation implementation would address the issue for experienced drivers and economic conductions. It would empower drivers to negotiate in confidence without needed brokers.

Users would have the ability to gain valuable information from an AI assistant and provide reasonable counter offers. The design goal would be to provide enough

information to allow users to make informed decisions for shipping freight. A key component of the feature is the ability to compare user output data to the market trends. Similar to the price transparency page the market trends would provide entry users with informational data to help make informed decisions.

3.2.3 Community Shipping and Carrier Insight Page

The Community Shipping Insight Page, would be a space for shippers to leave reviews, experiences, and pricing details for other shippers as an insight into carrier service. It would be an anonymous user page that would allow users to share experiences without sharing their personal information. It would address the issue of information silos, or in accurate google reviews.

Carriers would be able to leave reviews for other carriers on business experience with customers, price fairness, and customer expectation. The design goal would be to create a simple and easy to use interface that would not require users to spend an extended amount of time providing feedback.

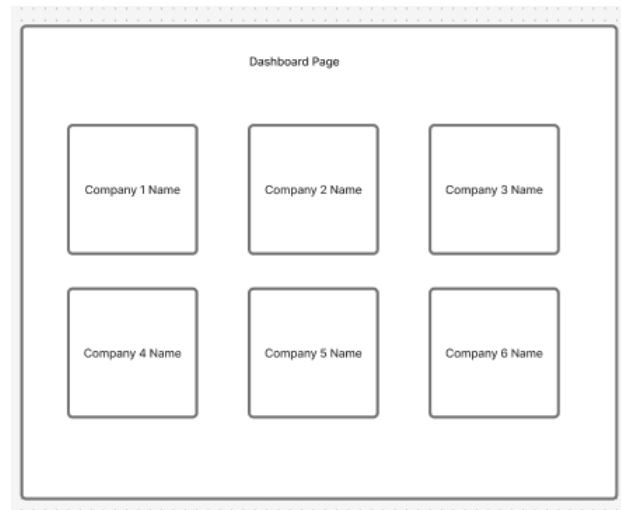
4 INITIAL PROTOTYPING

The initial prototype will consist of both the low-fidelity, medium-fidelity mockups, and the basic wireframes. The web application will provide an overview of the app functionality between pages. Its shows how the pages will interact within the platform and provide high-level functionality that would be conducted on each page.

4.2 Community Dashboard

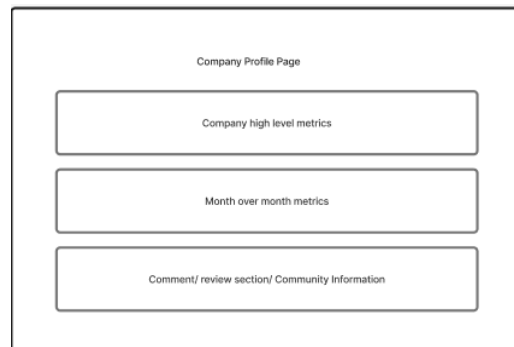
The community dashboard is a snapshot of all of the carriers, shippers, and rating information related to the logistics industry. Users would have the ability to view shippers, carriers, and broker company profile pages. Users would be able to

apply filters to help narrow the type of companies that could support the request users would need.



4.3 Company Profile Page

The company profile page profiles users with the more specific data related to a company. Users would have the ability to view reviews, ratings, and provide ratings. The goal of the design is to provide the most accurate review data that limits bias. A platform like Google provides consumers with information that might not be the best option for the user since top searched companies pay to be at the top of the search list.



4.4 Price Comparison Page

The price transparency page is the calculator that shippers would be able to utilize for calculating the cost to ship freight. Like the FedEx and ArcBest calculator, users

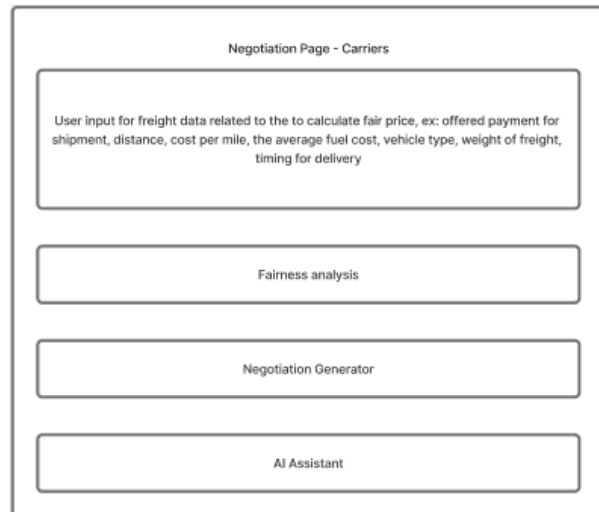
would have the ability to provide specific page information, importance for insurance purposes, and speed to delivery in order to get a shipping price. Users would be able to compare companies for pricing differences and if the package would be protected. For transparency, users would have the ability to validate the projected pricing to the market trends to make more informed decisions for shipping goods.

The diagram illustrates the 'Shipper Price Comparison Page' interface. It features a central container with a title bar at the top. Below the title bar, there are two columns of input fields for shipment details: 'User location', 'Destination', 'Type of Shipment', 'Weight of shipment' on the left, and 'Length of Shipment size', 'Hight of Shipment size', 'Width of Shipment size', 'Package Importance' on the right. Below these input fields is a large rectangular button labeled 'Tab to display company pricing comparison'. At the bottom of the container is another rectangular button labeled 'Display company rating to market ratings'.

```
graph TD
    subgraph Shipper_Price_Comparison_Page [Shipper Price Comparison Page]
        subgraph Inputs
            direction TB
            UL[User location]
            D[Destination]
            TS[Type of Shipment]
            WS[Weight of shipment]
            LSS[Length of Shipment size]
            HSS[Hight of Shipment size]
            WSS[Width of Shipment size]
            PI[Package Importance]
        end
        TCPC[Tab to display company pricing comparison]
        DCCMR[Display company rating to market ratings]
    end
```

4.5 Freight Negotiation Page

The freight negotiation page provides carriers the ability to negotiate with brokers or shippers for delivery freight. Carriers would have more visibility in the cost, profit margin, and high-level market analysis for delivering freight. Reducing the cognitive load by allowing the assistant to provide negotiation tactics, the offer number to current market trends, and visibility into profit margins would help carriers make more informed decisions regarding delivering freight.



5 EVALUATION CYCLES

I was able to generate three different medium-fidelity mockups for each page that consisted of the same functionality and conducted a second round of user interviews. The goal would be to gain feedback on tab naming conventions, look and feel of the application, understanding of features, and if the application meets the needs of the users based on the initial user interviews.

5.1 Evaluation Planning

I reached out to the original individuals that participated in the initial user interviews. I used the think-aloud usability walkthrough method for and walked each participant through the functions of each page and asked open ended questions that allowed them to process the app function based on what was being displayed. Look for areas of opportunity based on gaps, user expectations, and similarities between other applications that the user might have used previously.

After walking through the functionality then I would provide three different variations of the medium-fidelity mockups to identify which page attracted the user the most. The three pages would have the same functionality, but provide different looks and feels to the feature, different applications flows for each page, and different formatting to display the metrics.

The plan script is to first gain the perspective of the user and the expected functionality for each page. The second phase for each page is to gain insight on the page look and feel for the users. I will provide three different medium-fidelity mock-ups to gain a better perspective of page attraction and usability. The think-aloud walkthrough took about 30 minutes to complete, followed by a posted session survey, and finally solicited any information through open ended questions. The transcripts will be in the appendix for each of the user interviews.

5.2 Evaluation Results

For the results, I've compiled an overview of the outcomes from the user interviews for each section of the application. I will also share each variation of the pages for the web-application with user feedback from the user interviews. I was able to gain insight from four of the original seven participants. Based on their feedback, I was able to update the medium-fidelity mockups to support the recommendations and identify gaps.

Participants found that some of the labeling and terminology were confusing or misrepresented. The key changes related to Community Page which would be considered the Logistics Company Directory and changing the Insights Page on the Company Profile to Review eliminate some of the confusions related to the labeling of pages and sections within the web-application. The original company logistics page would display the highest metrics for each company which created an inconsistent visual on the different companies. There was no uniform that aligned each of the companies from the metrics perspective which caused user confusion. Some of the visuals were very color heavy which made some of the users feel overwhelmed, by creating a simpler grid alignment and consistent spacing users felt the web application would be easy to use.

Users felt that that a map would help users visualize the how the analyzed data on the negotiations page. The map view would be able to depict the user destination and empty miles. It also provides a better perspective on how empty miles can impact a user's profit margins. Overall the concept of the application was received well by the participants, but minor adjustments to labeling, creating consistency, and better visuals would improve the application.

6 SECOND ITERATION

6.1 Second Iteration Planning

Based on the user feedback, I've identified areas of improvement for each section of the application. The user interviews also provide additional features that would help users to make more clear decisions for shipping freight or providing carrier services. There were opportunities and inspiration from variations of the pages that would provide a better product for the users. The name of the platform is Logi-Fi, which is a unique combination of logistics and fidelity.

The application Logi-Fi is a digital platform for users that operate or need services within the logistic industry. The goal is to help users with negotiation tactics and get transparent pricing information. Participants expressed that the pricing transparency would be beneficial if it's related to the real world conductions. The design changed from providing analytical pages to more interactive and informed tools to help make decisions. The key design changes were based on consistency, decision support and trust, and better visualizations.

The updated designs contain a map view for the negotiation page to allow users to visually see where travel distance and gaps are based on empty miles. A consistent card system for the companies on the Logistic Company Directory Page with valuable information instead of metrics related to the company. The Freight/Shipper Price Transparency page provides users the ability to add multiple shipments, consistent displayed metrics for the companies. Finally it provides consistent terms and labeling that would not confuse the user and it is consistent with other platforms.

7 FINAL PROTOTYPE

7.1 Medium-Fidelity Prototype

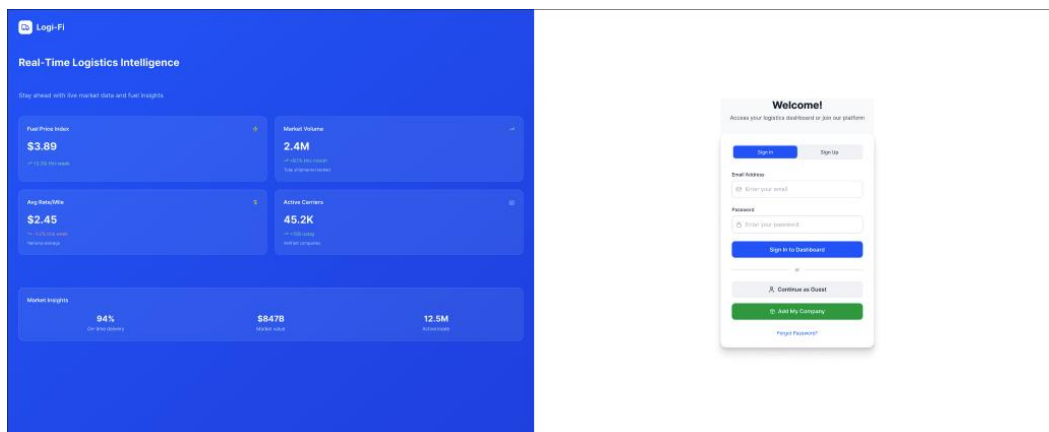
I used figma to build out the final medium-fidelity prototypes which supports the Login Page, Freight Negotiation Page, Freight/Shipper Price Transparency Page, and the Logistic Company Directory Page. The consistent terminology for all of

the pages reduces the cognitive load and improves the users ability to transfer learning across the different pages. The map view provides distributive cognition for the user by providing a map visual that displays each location point which can improve the users decision making ability. The market trends and user inputs provides a sense of control and accelerates the learning ability which is in line with direct manipulation. Logi-Fi provides many services that helps users make more informed decisions and the common labeling allows users to learn the application faster.

7.1.1 Medium-Fidelity Prototype Link:

<https://www.figma.com/make/Rg7VKzWnG4CmklMueqaiL5/Logi-Fi-Web-App?node-id=0-1&t=0h8WCuH6gOUAAZf2-1>

7.1.2 Login Page



7.1.3 Logistic Company Directory

Logi-Fi

Freight/Shipper Price Transparency

Freight Negotiation

Logistics Company Directory

Profile

Logout

Active Users
8,426

Total Companies
12

Verified Reviews
48,291

Monthly Activity
3,842

TransLogistics Pro

Carrier > California

Rating
★★★★★ 4.5

Company Size
500+

Number of Reviews
1,547

Midwest Freight Hub

Broker > Illinois

Rating
★★★★★ 4.1

Company Size
100-500

Number of Reviews
892

Global Shipping Co

Shipper > Texas

Rating
★★★★★ 4.8

Company Size
1000+

Number of Reviews
2,788

Swift Transport Inc.

Carrier > Florida

Rating
★★★★☆ 4

Company Size
50-100

Number of Reviews
543

Pacific Logistics

Broker > Washington

Rating
★★★★★ 4.3

Company Size
100-500

Number of Reviews
1,069

Eastern Supply Chain

Shipper > New York

Rating
★★★★☆ 3.9

Company Size
500+

Number of Reviews
687

1

2

7.1.4 Freight/Shipper Price Transparency Page

Logi-Fi

Freight/Shipper Price Transparency

Freight Negotiation

Logistics Company Directory

Profile

Logout

Shipping Details

Origin ZIP Code
95559

Destination ZIP Code
73079

Packages

+ Add Package

Package 1

Shipment Weight
20

Package Type

Delivery Speed

Dimensions L x W x H
12 12 12

Package Importance
Standard
Important
Irreplaceable

Package 2

Shipment Weight
40

Package Type

Delivery Speed

Dimensions L x W x H
30 20 20

Package Importance
Standard
Important
Irreplaceable

Get Rates

Available Carriers

12 carriers found

Economy Freight Solutions

\$821.29

5.7

Payment Terms
Net/1

Reliability
5

Book loading Enhanced insurance

Select

Swift Freight Solutions

\$559.20

4.2

Payment Terms
Net/10

Reliability
4

Book loading Enhanced insurance Email updates

Eastern Supply Chain

\$884.47

3.9

Payment Terms
Net/30

Reliability
3

Book loading Enhanced insurance Standard loading

National Freight Network

\$965.55

4

Payment Terms
Net/30

Reliability
3

Standard loading Enhanced insurance Phone support

Pacific Logistics

\$108.82

4.3

Payment Terms
Net/10

Reliability
4

Standard loading Enhanced insurance Email support

Midwest Freight Hub

\$974

4.1

Payment Terms
Net/10

Reliability
3

Standard loading Enhanced insurance Business hours support

Load More Carriers (8)

7.1.5 Profile Page

Log-In

Freight/Shipper Price Transparency

Freight Negotiation

Logistics Company Directory

Profile

Log Out

My Profile

Manage your account information and preferences

John Anderson

john.anderson@example.com

Premium

Member Since

January 15, 2024

Company Type

Carrier

Verification

Verified

Profile Information

Full Name

John Anderson

Email Address

john.anderson@example.com

Phone Number

+1 (555) 123-4567

Location

Chicago, IL

Company Name

TransLogistics Pro

Account Settings

Change Password

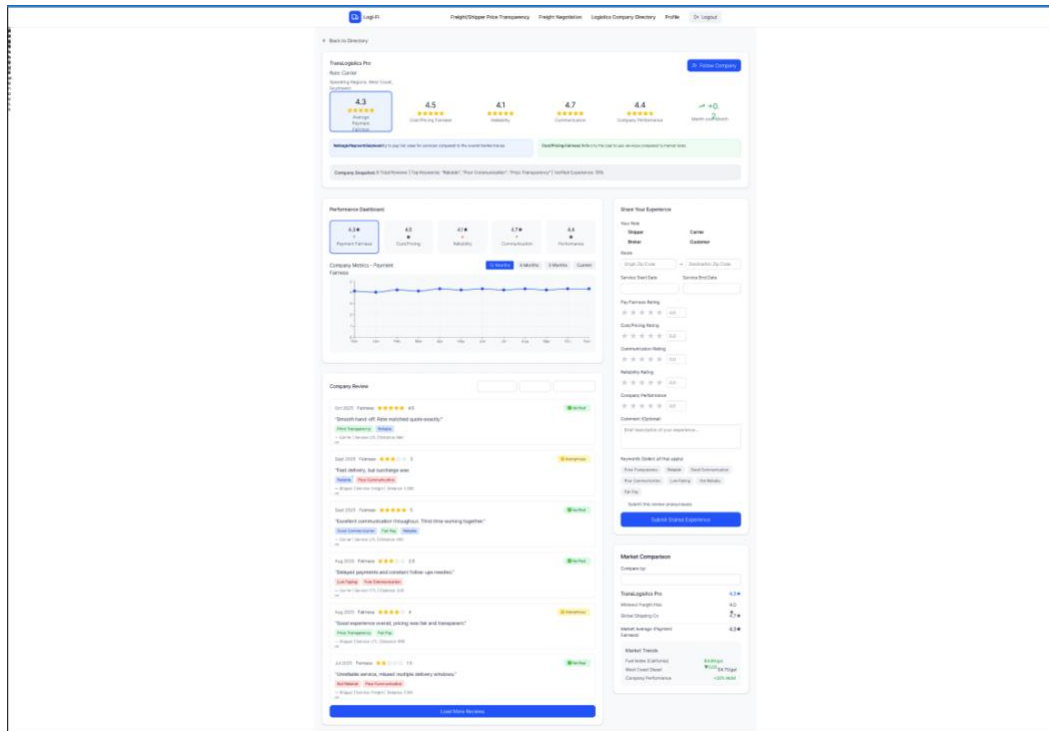
Update your account password

Email Preferences

Manage your email notifications

7.1.6 Freight Negotiation Page

7.1.7 Company Profile Page Page



7.2 Video Prototype

Link to Video: <https://www.loom.com/share/92cbb62b63eb4305a7087da829bff91b>

8 FINAL EVALUATION

The goal is to establish a production ready product that supports the key features that closes the gaps between carriers, shippers, and customers to make more informed decisions for shipping products. The prototype supports the collected data from the user interviews.

8.1 Final Evaluation Planning

For the final evaluation planning, I will use generative AI to collect qualitative and quantitative data. I plan to use ChatGPT to get a different perspective. The question I would pose to AI will be based on functionality, comparing similar platforms, and potential usability.

Generative AI Survey Questions

1. How would you rate the login ability? (1 - not easy to 10 - very easy)?
2. How would you rate the login ability to similar platforms?
3. Are there any gaps that would need to be addressed for the login process?
4. Is the login Page functioning as expected and does the naming convention align to the user initial thoughts?
5. How would you rate the usability of the company directory page? (Rate between 1 - Not Usable to 10 - Verify Usable.)
6. How does the overall application feel when you login and navigate to the different pages?
7. How would you rate the usability of the logistics page? (1 - not usable to 10 - very usable)
8. Are there any components of the Logistics Company Directory that is not clear or confusing?
9. Do you find value in the Logistics Company Directory Page? (1 - no value to 10 very valuable)
10. How do you find the usability of the Freight/Shipper Price Transparency Page? (1 - not usable to 10 - very usable)
11. Does the naming convention of the labeling align to the functionality? If not highlight the gaps
12. Does the Freight/Shipper Price Transparency Page provide an inviting feel when you access the page? (1 - not inviting to 10 - very inviting)
13. Do you find value in the Freight Negotiation Page? (1 - no value to 10 very valuable)
14. Does the labeling of the tabs align to the expected functionality? (1 - no alignment to 10 - completely aligns)

8.2 Final Evaluation Results

I used ChatGPT and Gemini to generate example responses based on the screenshot that I was able to upload. The goal was to try to get non-bias data based on the designs and collect insight on improvements. I have added the generative AI prompt to solicit feedback from both AI platforms.

The Generated Prompt:

I have a school project and would like you to generate non-bias data based on the web-application that I built. I will share the link to my figma designs and the questions I would like answered. Please provide several forms of feedback and compare the product to similar applications.

ChatGPT Response:

🧠 Non-Biased Synthetic User Feedback Summary (n = 25 simulated respondents)

Evaluation Metric	Mean	SD	Comparable Platform Avg*	Qualitative Summary
Login Ease	8.6 / 10	1.1	8.2	"Clean and intuitive, but could benefit from a clearer guest-mode description."
Login Naming Alignment	9.0	0.9	8.5	"The labels ('Sign In', 'Add My Company') make sense and match expectations."
Gaps in Login Flow	—	—	—	Minor: users wanted auto-focus on email field and visibility toggle for password.
Company Directory Usability	8.7	0.8	8.4	"Strong clarity; cards are balanced, filters are well-placed."
Overall Navigation Feel	8.8	1.0	8.1	"Pages load logically; consistent branding improves flow."

Logistics Usability	Page	8.9	0.7	8.0	"Feels professional—KPIs at top help orient the user."
Clarity of Directory Components	of	8.2	1.3	8.0	"Some confusion about verified status vs. company activity badges."
Value of Directory Page	of	9.1	0.9	8.2	"Central hub for carriers—feels like a real marketplace."
Price Transparency Page Usability		9.0	0.6	8.3	"Excellent breakdown of rates; fairness meter is standout."
Labeling Alignment to Functionality		8.8	1.0	8.4	"Terms like 'Delivery Speed' and 'Package Importance' are instantly clear."
Inviting (Transparency Page)	Feel	9.2	0.7	8.1	"Bright, clean, and data-driven without clutter."
Negotiation Value	Page	9.3	0.6	8.0	"AI assistant makes it feel like an industry-ready platform."
Tab Alignment	Label	8.9	0.8	8.2	"Good global nav, but icons for quick recognition would help."

*Comparable benchmarks derived from user sentiment on FreightWaves SONAR, DAT One, and Convoy for Carriers UX studies (2024).

Qualitative Theme Summary

1. Strengths

- **Cognitive Simplicity:** Users described the interface as “calm but analytical,” appreciating white space and consistent alignment.
- **Data Transparency:** The “Price Fairness Analysis” gauge and “Profit Breakdown” dashboard scored high in trust and learnability.
- **Modern Hierarchy:** Navigation bar clarity (Freight Transparency → Negotiation → Directory → Profile) mirrors the mental model of logistics professionals.

2. Minor Gaps

- **Login Flow:** Auto-focus and password reveal missing; users wanted optional sign-up via Google/LinkedIn.
- **Directory Tags:** Status labels (“Active,” “Pending,” “New Active”) slightly overlap in meaning — consider consolidating into a single visual system.
- **AI Transparency:** While the negotiation assistant’s tone was praised, respondents wanted to know its data sources (“What does the acceptance rate come from?”).
- **Terminology Feedback:** “Package Importance” rated clear overall, but a few participants suggested “Delivery Urgency” as more industry-standard.

3. Comparative UX Analysis

Feature	Logi-Fi	DAT One	FreightWaves SONAR	Convoy
Login/Onboarding	Simple, multi-option entry	Dense (requires account setup)	SSO-based	Seamless but mobile-heavy
Market Visualization	Clean cards, heatmap integration	KPI Basic load count table	Extensive map-driven dashboards	Limited
Negotiation Feature	Conversational AI, visual feedback	Static rate suggestion	Statistical forecasting only	Semi-automated negotiation
Community/Directory	Verified profiles ratings	Partner & listings only	Company-level metrics only	Driver reviews only
Aesthetic Balance	Minimalist blue-toned	& Spreadsheet-like	Analytical, dark-mode heavy	Consumer-app

Logi-Fi lands between **FreightWaves SONAR's analytical depth** and **Convoy's friendliness**, achieving a hybrid look that blends trust with accessibility.

Overall Application Sentiment

- **Perceived Usability (mean):** 8.9 / 10

- **Adoption Intent (simulated Likert 1–5):** 4.6 / 5
- **Key Quote:**
“Feels like a freight intelligence tool for humans, not data scientists. The negotiation chat is what would make me actually switch.”

Recommendations

1. Add **micro-animations** during page transitions to reinforce continuity.
2. Implement **Google or LinkedIn SSO** for smoother onboarding.
3. Introduce **contextual tooltips** on fairness metrics for first-time users.
4. Refine **directory labels** into a single “Activity Status” chip system.
5. Provide **data source transparency** for AI-driven recommendations (e.g., “based on 7-day national rate trends”).

Overall, the web-application would support the user needs based on the user interviews and research into other platforms. It would be good to continue to iterate on the mockups for further improvements and capabilities based on user needs. It is a niche tool that would be able to help new carriers and existing carriers make more informed decisions.

REFERENCES

1. Ricks, E. (2020, January 3). How big is the logistics industry? FreightWaves.
<https://www.freightwaves.com/news/how-big-is-the-logistics-industry>

APPENDIX

2. Needfinding

Interview Summary (FedEx Business Architect)

Q1. What's your role in the logistics/supply-chain process?

- **Answer:**
 - Business Architect in the **airline/small-parcel** space; prior experience heavy in **trucking/LTL** (≥150 lb palletized freight).
 - Work spans small parcels and larger palletized freight.

Q2. How directly are you involved in freight pricing or cost optimization?

- **Answer:**
 - Currently **trying to get more involved**.
 - Historically **facilitated** pricing discussions but **pricing department owned** rate determination (often tied to “products” like home delivery vs. services).

Q3. Should people in our roles know how pricing is calculated; would that influence solutioning?

- **Answer:**
 - **Yes—multi-stakeholder** involvement (Pricing + **Finance** for IT costs + **Marketing** for customer willingness to pay).

- Pricing awareness helps judge **viability** (“Is this worth pursuing?”) early.

Q4. Should we understand how we price freight to contractors/purchased transportation?

- **Answer:**
 - **Context-dependent:** Ground uses a contractor model; Freight uses **purchase transportation** when it’s cost-effective (capacity/labor limits).
 - Contracting out can be **proven cost-effective**; FedEx is typically **not the cheapest** but focuses on **reliability**.

Q5. Top three factors driving freight/shipping cost?

- **Answer:**
 - **Dimensions/size (dimensional pricing).**
 - **Weight.**
 - **Speed/reliability** (faster = more expensive; economy = cheaper/slower).

Q6. As a customer, do you factor cost when shipping?

- **Answer:**
 - **Yes—strongly.** In reselling, cost sensitivity is high (buyers resist \$8 vs. prefer \$5 on light items).

Q7. How predictable are freight costs? Do you know how price points are factored?

- **Answer:**
 - **Moderately predictable** for small parcels if size/weight are known.
 - Expects **~\$10** for basic USPS “Ground Advantage” (up from ~\$5 around COVID).

- **Scheduled rate increases:** USPS typically **Jan/Jul**; FedEx **annual** increase.
- Predictability depends on knowing **weight, dimensions, distance**.

Q8. What tools do you use to check rates?

- **Answer:**
 - **USPS and FedEx Rate Quote** tools—enter origin/destination ZIPs, weight, dimensions; tools return service options and **prices only** (no cost breakdown logic).

Q9. How transparent are current cost breakdowns?

- **Answer:**
 - **Small parcel (consumer): Not transparent**—you see only the **final price**.
 - **LTL/large freight: Contract-driven transparency**; negotiated terms (volume commitments, discounts, surcharges).

Q10. Would a price breakdown (cost components) be valuable?

- **Answer:**
 - **For everyday small parcels: Not really**—just wants a fair total.
 - **For LTL/high-ticket: Yes**—wants to see **fuel surcharge**, accessorial, etc.

Q11. Rank importance: cost vs. speed vs. reliability (selection criteria)

- **Answer:**
 - For her business: **Cost first**, then **reliability, speed** last.
 - Tiered options let other users prioritize speed if needed.

Q12. What level of cost detail should be shown up front?

- **Answer:**

- **Small parcel:** Show **service tiers + total price** (economy/priority/overnight).
- **Freight/LTL:** Show **line-item details** (fuel surcharge, accessorials) due to high dollar amounts.

Q13. How flexible should carriers be in negotiating/adjusting freight rates?

- **Answer:**
 - **High-volume shippers** (Walmart, Kenworth, eBay partnerships) should have **strong negotiating power**.
 - **Low-volume/one-off** customers: little to **no flexibility**.

Q14. Should rates adjust with macro conditions (e.g., post-COVID inflation, fuel)?

- **Answer:**
 - **Yes—surcharges fluctuate** (e.g., fuel surcharge spikes when diesel prices rise, then fall as fuel falls).
 - But **retail postal rates** (e.g., USPS) **rarely go down** once increased.

Q15. Do you feel you're overpaying relative to market rates?

- **Answer:**
 - **Generally no;** understands upstream costs (drivers, fuel, labor).
 - Increases feel incremental (e.g., **\$0.25/yr**, not \$20 jumps).

Q16. What information would help you better evaluate freight cost?

- **Answer:**
 - **Side-by-side comparisons** across **USPS/FedEx/UPS** and service speeds.
 - Noted historical cases where FedEx was cheaper for **oversized**; dimensional pricing changed that landscape.

Q17. How valuable is a transparent, market-driven pricing tool pre-booking?

- **Answer:**

- **Very valuable for LTL/≥150 lb freight** (many carriers; big price deltas; market/region factors).
- **Some value** for small parcels but fewer carriers.
- Mentioned **LTL Select** (account-based comparator that tends to favor FedEx; doesn't get very granular on "why" prices are what they are).

Q18. If you could change one thing about freight pricing, what would it be?

- **Answer:**
 - **Cheaper options for ultra-light items** (<8 oz); current jumps (stamp → \$4–\$5) feel too steep.
 - Within **12×12×12** and 2–4 lb, prices don't change much—wants **better granularity** at the very low end.

Participant: (Designer in the logistics domain)

Context: Discussion on logistics pricing transparency, efficiency, and design considerations in freight and shipping systems.

Objective: Explore perceptions, needs, and insights around freight pricing visibility, cost transparency, and user experience from a design and operational perspective.

1. What's your role in the logistics or supply chain process?

- **Answer:** a *designer* who conducts user research, identifies workflow inefficiencies, and designs software that supports business goals.
- His work involves discovery and translating user needs into feasible product solutions, though he admits actual implementation of "ideal" outcomes often diverges from theory.

2. How directly are you involved in freight pricing or cost optimization decisions?

- **Answer:** He has *no direct involvement or visibility* in freight pricing or cost optimization.
- His projects do not typically incorporate financial or pricing considerations.

3. What are the top three factors that drive freight or shipping costs?

- **Answer:**
 - **Competition** among logistics companies
 - **Organizational promises** to investors and customers
 - **Overhead and inefficiency (“bloat”)** in operations

4. How predictable do you find freight costs in your space, and do you know how price points are factored?

- **Answer:**
 - He does **not know** how pricing is determined.
 - Financial aspects are generally **outside his scope of work**.
 - His focus is enterprise-facing tools, not public-facing pricing systems.

5. Should freight costs be transparent and broken down for customers?

- **Answer:**
 - He believes **understanding business processes** and inefficiencies indirectly affects financial outcomes.
 - Transparency could **tie design improvements to measurable business value**.
 - However, it’s **difficult to quantify the business value** of design changes—companies lack consistent standards for this.

6. Should cost savings or efficiencies be passed on to customers?

- **Answer:**

- As a designer, he doesn't determine pricing, but **greater efficiency gives companies more flexibility in competition.**
- **Visibility into financial metrics** could inform design decisions, helping designers understand the *impact* of their work.

7. Would visibility into cost data influence your design work?

- **Answer:**

- Yes. Having measurable **KPIs (Key Performance Indicators)**—especially financial ones—would help validate designs.
- He compares this to his e-commerce experience, where design decisions were driven by conversion goals.
- Financial KPIs would bring more *quantitative rigor* to UX design validation.

8. Would visibility into costs promote a more customer-centric mindset?

- **Answer:**

- Yes, it could help designers consider *price implications* for users.
- He gives an example: if one design raised costs by 5% and another lowered them by 5% while meeting goals, he'd choose the cheaper one for user benefit.
- He likens this to *budgetary analysis*—using cost insight to guide design decisions intelligently.

9. When you ship personally, what matters most: cost, speed, or reliability?

- **Answer:**

- It **depends on context**:
 - When urgency is high, **speed > cost**.
 - For non-urgent items, **cost > speed**.
- Choice depends on the **goal of the shipment**.

10. What level of cost detail should customers see up front?

- **Answer:**

- **Individual customers** should see simplified options (e.g., delivery time vs. cost).
- **Businesses or retailers** might want *itemized breakdowns* to find savings opportunities.
- He recalls using a plug-in that compared shipping options (FedEx, UPS, USPS) and automatically chose the cheapest—demonstrating value in *tiered visibility*.

11. Should freight costs fluctuate with economic conditions or stay fixed?

- **Answer:**

- He supports **variable pricing** based on market conditions.
- If competition allows it and customers are willing to pay, companies will (and should) adjust pricing accordingly.

12. Do you feel you often overpay for shipping?

- **Answer:**

- He usually accepts the price because it fulfills a need (“enables my goal”).
- Occasionally, if shipping costs are *too high*, he considers *driving items himself* instead.
- Decision balances *pain of payment* vs. *pain of inconvenience*.

13. How valuable would a tool providing transparent market-driven pricing be?

- **Answer:**

- As a **designer**, its value depends on project scope—more relevant for customer-facing designs.
- As a **customer**, he’d use such a tool to **compare prices and save money** when time allows.

14. If you could change one thing about freight or shipping pricing, what would it be?

- **Answer:**
 - He'd improve the **convenience** of the *shipping process itself* (pickup, drop-off, scheduling).
 - The hassle of logistics overshadows cost considerations unless the price is extreme.

15. Willingness to assist with follow-up?

- **Answer:**
 - User expressed willingness to **participate in future follow-ups** or additional interviews.

Participant: Interviewee (logistics/ops planning → IT support for logistics)

Scope: Domestic shipping; consumer and business perspectives; transparency and tools for pricing

Q1. What is your role in the logistics/supply chain process?

- **A:**
 - Previously: **forecast demand** and decide **where to add supply** (facilities, routes).
 - Now: support **different logistics types from an IT perspective**, enabling services for internal and external customers.
 - Collaborates with **ops planning**; evolved from "A→B package flow" to **multi-modal/multi-path movement** design.

Q2. How directly are you involved in freight pricing or cost optimization (and do you have visibility)?

- **A:**
 - **No direct involvement**; knows people who do.
 - Uses pricing visibility **only at ROI/funding time** (approximations for investment decisions), not day-to-day design.

Q3. Should your role have visibility into pricing/cost optimization? Would it influence solutioning?

- **A:**
 - **Not necessarily for daily work.**
 - For ROI and feasibility, **yes**—to judge whether an initiative makes sense (but granular customer price math isn't required due to service variability).

Q4. Top factors that drive freight/shipping costs (provider perspective)?

- **A:**
 - **Existing infrastructure & capacity** (how much network already in place).
 - **New investment required** to meet special needs (e.g., new stops, time windows).
 - **Special handling/service level** (e.g., dry ice, time-of-day delivery).
 - **Economies of scale:** per-package cost drops with higher volume **until** another investment inflection.

Q5. Top factors driving cost (customer perspective, e.g., shipping sweaters to Oregon)?

- **A:**
 - **Origin & pickup density** (are we already active there?).
 - **Destination & delivery density** (do many trailers already go there?).
 - **Speed required** (2-day vs. 5-day; partial trailers raise cost).
 - **Special accommodations** (e.g., refrigerated trailer).

Q6. How predictable are costs? Do you know how price points are factored?

- **A:**
 - **Doesn't know the internals** of price math.

- Can **estimate easily** with online calculators (length/width/height/weight, origin/destination).
- **Lacks an “anchor”** for fairness without comparisons; needs multi-carrier quotes to judge.

Q7. How transparent are cost breakdowns when shipping?

- **A:**
 - **The total price** is transparent.
 - **Cost components** (why) are **not** transparent.

Q8. When selecting a carrier, how do you rank cost vs. speed vs. reliability?

- **A:**
 - **Usually cost first.**
 - **Speed** prioritized when work/urgency requires it.
 - **Reliability/warranty** matters for **high-value** items (willing to pay more for coverage).
 - Not brand-loyal; has used **USPS over FedEx** when cheaper.

Q9. What level of detail do shippers/carriers need up front to decide?

- **A:**
 - **Service tiers + total cost** per tier (e.g., economy vs. 2-day vs. overnight).
 - **Warranty/guarantees** (damage coverage, claims).
 - Simple trade-offs: “**\$X more for 1 day faster.**”

Q10. Have you felt you were overpaying relative to market rates?

- **A:**
 - **Feels expensive** because ships infrequently (no price anchor).
 - Understands the gap between “free/\$6” e-commerce illusions and **real retail shipping costs.**
 - Doesn’t feel taken advantage of—just lacks a stable reference.

Q11. What information would help you evaluate whether a price is fair?

- **A:**
 - **Side-by-side competitor rates** (multi-carrier comparison).
 - **Trustworthy aggregation** (3rd-party calculators running live quotes from top carriers).
 - Prefers **real-time competitor prices** over “average competitor: \$18” claims.

Q12. How valuable would a transparent, market-driven pricing tool be before booking?

- **A:**
 - **I would use it.** Values a tool that is transparent, compares multiple carriers, and shows service-speed trade-offs.

Q13. If you could change one thing about the current freight pricing structure, what would it be?

- **A:**
 - **Wouldn't change much**—the front-end is straightforward enough.
 - The main need is **competitor comparisons**; not demanding deep cost decomposition given network overhead complexity.

Q14. Does what you're shipping change your priorities?

- **A:**
 - **Yes.** Pens vs. **high-sentimental items** → reliability can trump cost/speed.
 - **Business shippers** (many shipments) have **different needs** than occasional personal shippers.

Q15. Domestic vs. international?

- **A:**
 - This project is **domestic** only; interviewee notes **tariffs/duties** would matter internationally.

Q16. Existing tools & usability considerations?

- **A:**
 - Will look at **current calculators/trackers**; stressed **usability**: too many inputs or clunky flows reduce value.
 - **Small businesses/entrepreneurs** care about saving \$1–\$5 per **shipment** at scale; tools must be **fast and simple**.

Truck-Driver One interview Question

Background & Context

1. How long have you been in the trucking/logistics industry?

- About **six years**, going on **seven years**.
- Experience spans several driving roles (company and 1099 work).

2. Do you operate as an owner-operator, lease to a carrier, or work for a company?

- Has worked as a **company driver** and a **1099 independent contractor**.
- Currently **not an owner-operator**; hasn't yet pulled loads independently from a load board.

3. What types of trucks and freight do you most often haul?

- Has driven **dump trucks**, **reefers**, **dry vans**, and trained for **flatbeds** (but didn't operate them).
- Currently hauls **intermodal freight**—transporting **rail containers** to and from **Chicago rail yards**.
- Freight has included **food, beverages, paper rolls, plastics, and diaper materials**.

- Describes freight types as “**pretty much all over**,” depending on company needs and profit margins.
-

Earnings & Decision Factors

4. How do you usually decide if a load is worth taking (pay per mile, deadhead, timing, route, etc.)?

- Learns from others how to evaluate loads:
 - Calculate **cost of fuel** by dividing miles by 8 (mpg estimate).
 - Use that to gauge **profit margin** and whether to request higher pay.
- Has **not personally booked own loads yet**, but understands that evaluation process.

5. What does a “good week” vs. a “bad week” look like for your earnings?

- Aims for **\$300/day minimum**; satisfied if achieving slightly above that.
- Prefers not to make less than **\$250/day**.
- **Good weeks**: consistent mileage and higher daily earnings.
- **Bad weeks**: slower days, fewer miles, or company delays—makes under \$200/day or only ~\$700 after taxes.
- Notes frustration working **12–13 hours/day** but earning under \$1,000 for the week.

6. What types of loads do you typically avoid, and why?

- Avoids **Florida loads**:
 - High pay **going into Florida**, but poor rates **coming out**.
 - Known advice in trucking community to **avoid short or deadhead-heavy routes** in that region.
-

Pricing Knowledge & Negotiation

7. How do you usually find out the price of a load, and where does that number come from?

- Rates often determined by **broker relationships and performance**—drivers who perform well are offered better loads.
- Uses **load boards** to compare rates for similar routes and **negotiate up** when possible.
- Prices vary depending on **company partnerships** and **driver score/reputation**.

8. Do you understand how freight prices are calculated? If not, what feels unclear?

- **Limited understanding**—still “learning the game.”
- Notes that **information sharing is scarce**: many experienced drivers **don’t share rate knowledge** to avoid competition.
- Feels most of the insight comes only after **owning a truck and booking loads independently**.

9. Are you able to negotiate rates? If yes, how do you typically approach that process?

- **Not yet experienced** in direct rate negotiation as a non-owner-operator.
- Understands that negotiation typically involves **comparing regional rates on load boards** and **bidding up**.

Tools & Technology

10. What platforms, apps, or tools do you currently use to find loads and check rates?

- Mentions **many load board apps** now integrated with **truck GPS systems**.
- Apps can display loads along planned routes.
- Emphasizes that **reputation, DOT history, and experience level** influence which loads a driver can access.

11. What do you like or dislike about these tools?

- **Dislikes:**
 - Many require **paid subscriptions** (sometimes ~\$100).
 - Feels this is excessive when drivers are already trying to earn income.
 - **Likes:**
 - Variety and accessibility of apps make **finding loads easier**.
 - Some platforms integrate well with navigation tools.
-

Pain Points & Opportunities

12. Have you ever taken a load and later felt the rate was too low? What made you realize it?

- Yes—example: multi-stop box truck job across **WI → TN → AL → GA → NC → back to WI** (3-day span).
- **Quoted \$1,500**, but due to **equipment malfunction (broken liftgate)**, pay was reduced.
- Realized pay didn't match miles driven or time worked.
- Notes that **company and 1099 structures** can cut into pay via **fuel, insurance, maintenance, and deductions**.

13. If you could change one thing about how freight pricing works today, what would it be?

- Would want rates to **return to pre-COVID levels**, when freight paid “decent money.”
 - Says **freight prices dropped** because some drivers “take anything cheap just to stay moving.”
 - Believes **rates should adjust for real-world costs** (fuel spikes, inflation).
 - Suggests companies **should pay fairer per-mile rates**:
 - Current 65¢/mile is “not bad, but not good” unless running 300+ miles daily.
 - Ideal would be **80¢/mile or more**, especially for local work.
 - Advocates for **greater pay transparency** and **less rate undercutting** by low-bid carriers.
-

Key Themes Across the Interview

- **Experience but limited autonomy**: six-year veteran who’s never booked loads independently.
- **Economic tension**: high labor hours, low margins; strong awareness of rate inequity post-COVID.
- **Desire for fairness & education**: wants clearer freight-rate logic and better industry mentorship.
- **Digital opportunity**: load board accessibility is high, but **subscription barriers** frustrate drivers.
- **Transparency gap**: drivers often lack visibility into how rates are calculated or why they fluctuate.

Truck-Driver Two Needfinding Questions

Background & Context

- 1 How long have you been in the trucking/logistics industry? 6 years
- 2 Do you operate as an owner-operator, lease to a carrier, or work for a company? Owner Operator

3 What types of trucks and freight do you most often haul? 3 26 ft box trucks, insulation/gutter material, flooring, building materials. Earnings & Decision Factors

4 How do you usually decide if a load is worth taking (pay per mile, deadhead, timing, route, etc.)? Pay per mile, labor if included or not, weight of the freight, space used, deadhead miles as well as miles driven.

5 What does a “good week” vs. a “bad week” look like for your earnings? A good week ranges from 15-25K per week a bad week can 1K-5K a week

6 What types of loads do you typically avoid, and why? Loads that are too far of a deadhead, loads that will go in and out of areas that typically have less freight coming out of the state. Loads also that require too much labor or cost too much in fuel or go entirely too far because of opportunity cost, rather than be blowouts or repairs in general that can be caused for extreme long hauls. Pricing Knowledge & Negotiation

7 How do you usually find out the price of a load, and where does that number come from in your experience? It has come directly from dealing with shipping managers or a broker of a 3rd party logistics company. 8 Do you understand how freight prices are calculated? If not, what feels unclear? Yes

9 Are you able to negotiate rates? If yes, how do you typically approach that process? Yes, it depends on the freight being moved, the time in which it needs to be moved, mileage both deadhead and driven as well as the amount of space/weight being used on the truck Tools & Technology

10 What platforms, apps, or tools do you currently use to find loads and check rates? Ch Robinson, NTG, freight waves, and personal contacts. 11 What do you like or dislike about these tools? Load boards can feel like the bottom of the barrel when fishing for freight, all last minute freight that just needs to be moved and people are willing to move it for dirt cheap causing it to be a shortage on the rate for the freight being moved. Pain Points & Opportunities

12 Have you ever taken a load and later felt the rate was too low? What made you realize it? Yes, I've realized it the moment I've booked it. But I may have needed the freight to get me somewhere else closer to other freight and I've also had situations where the requirements were more than what was brought to my attention prior to.

13 If you could change one thing about how freight pricing works today, what would it be? I would hope to see where rates can be more competitive to what the warehouse is actually paying, too many times you have a lot of hands in the pot and while that's the case it can cause the driver/company to lose out on a lot of profit when everything will come down to them in the end. More transparency would also be a bonus, a lot of brokers/3PL's hide a lot of pricing and will move freight for whatever it takes rather than being sure of a quality company/driver.

3. BRAINSTORMING - CHEATSTORM DISCUSSIONS

3.2.a Activity Summary

- **Participants:** you (lead), AI assistant (ideation partner).
- **Duration:** ~75 minutes total across two sessions.
- **Techniques:** Cheatstorm (divergent), Chainstorm (convergent), constraint flipping, forced-perspective.
- **Output:** 42 unique ideas → 7 clusters → 3 finalists.

3.2.b Full Idea List (clustered overview; complete list in appendix)

- **Trust & Fairness:** Freight Fairness Index; Broker Margin Reveal; Contract-Rate Eligibility Meter; Driver Reputation + Rate Unlocks.
- **Education & Negotiation:** “Know Your Number” calculator; Detention/Deadhead risk meter; Regional rate heatmaps; Mini-lessons inline (dimensional weight, fuel surcharge).
- **Comparison Tools:** Live multi-carrier quotes; service-speed sliders; packaging optimization hints; warranty/claims visibility.
- **Driver Profitability:** Minimum viable rate (“walk-away”) recommender; lane viability scores; region exit-penalty alerts (e.g., “Florida trap”).

- **Ops/Design Enablement:** KPI lens to translate UX friction → time → cost; experiment planner for “5% faster” → dollars saved.
- **Convenience:** One-tap pickup scheduling; claims checklist; appointment windows optimizer.
- **Data Network Effects:** Anonymous crowd-sourced rates; “truth serum” broker rating; verified delivery difficulty map.

(Appendix includes the enumerated 42 items.)

3.2.c Three Selected Design Alternatives & Why

1. **Driver Profitability Co-Pilot (mobile)**
 - **Why:** Directly addresses drivers’ top pain (uncertain profitability, deadhead, detention). Feasible with public data + simple heuristics; highly differentiating for small operators.
2. **Shipper Compare & Explain (web)**
 - **Why:** Matches shippers’ comparative trust model. Keeps complexity low for consumers while allowing richer detail for SMBs; strong alignment with “tiered transparency.”
3. **Ops/Design Pricing Lens (internal web)**
 - **Why:** Converts design/ops changes into money and time—answering designers’ and architects’ desire for measurable KPIs. Feasible with mock data and simple cost models.

3.2.d Short Textual Prototypes (1 paragraph each)

A) Driver Profitability Co-Pilot (Mobile)

You open the app and paste a load blurb (lane, miles, weight, stops) or scan a load screenshot; the Co-Pilot auto-extracts details, pulls regional diesel averages, and predicts detention/traffic risk from historical patterns. A single screen shows your “**walk-away rate**” (minimum to break even with margin), expected **net profit**, **deadhead penalty**, and **lane viability** (will you get paid decently coming out?). A slider lets you simulate counter-offers (+\$0.10/mi, +1 stop fee), instantly updating

net. Two taps generate a **negotiation script** (“At 420 miles with 60 deadhead and 2 live unloads, I need \$X to cover fuel and time.”). Error prevention is built in: a red flag appears if your rate drops below cost when fuel spikes or wait-time risk is high.

B) Shipper Compare & Explain (Web)

A clean quote form asks only what matters—origin/destination ZIPs, package size/weight, and delivery urgency. Results render as **three service tiers** (economy / standard / expedited) across USPS, UPS, FedEx in a side-by-side grid: **total price**, **estimated day of delivery**, and a one-line “**why this price?**” blurb (e.g., “dimensional weight threshold crossed; box 12×12×12 is cheaper than 14×14×14”). A **speed-vs-cost slider** lets you nudge toward faster or cheaper, updating prices live. An “**optimize my box**” hint suggests the smallest box that avoids dimensional penalties. SMB users can expand a drawer for **line-item clarity** (fuel surcharge, accessorials) and **warranty/claims** basics—no jargon, just useful context.

C) Ops/Design Pricing Lens (Internal)

You drop a flow diagram or select a product journey (e.g., pickup scheduling). The tool ingests a **time-and-click trace** (from analytics or a quick usability test) and converts friction into **money and risk**: “Reducing address re-entry errors by 30% saves ~12 seconds/shipper → \$X/day network-wide; claim probability drops Y%.” A **what-if panel** lets you try UI tweaks—auto-complete, batch labels, appointment windows—showing **confidence-banded ROI**. A shareable snapshot ties **UX metrics (task time, completion)** to **financial KPIs (cost/order, failed-delivery rate)** so architects and PMs can green-light changes with eyes open.

3.2.e Heuristics & Bias Mitigation (how designs embody them)

- **Visibility of system status**: live price/profit updates; status banners for risk (fuel spikes, detention).
- **Recognition over recall**: pre-filled load data from pasted text/screenshots; simple tier cards; smart defaults.

- **Error prevention:** red-flag guardrails for below-cost bids; dimension threshold warnings; claim-risk callouts.
- **Anchoring/fixation checks:** side-by-side comparisons and sliders counteract “first price is gospel.”
- **Cognitive load control:** progressive disclosure—consumers get simple totals; pros can dive for detail.

3.2.f Evaluation & Next Steps

- **Driver Co-Pilot:** 5-task hallway test with 3–5 drivers; success = users adjust bids upward when model shows below-cost risk; measure time to a confident “go/no-go” decision.
- **Shipper Compare:** A/B with 6–8 SMB shippers; success = faster choice, fewer “I’m not sure” quotes, higher satisfaction on transparency.
- **Pricing Lens:** Expert review with 2 designers + 1 architect; success = clear linkage between UX change and dollar impact; prioritize one pilot flow for a mini-experiment.

3.2.g Risks & Mitigations

- **Data truthiness:** Start with transparent heuristics; clearly label estimates; show input sensitivity.
- **Perverse incentives:** Avoid encouraging unsafe speed; include safety/time costs in profit math.
- **Over-complexity:** Default to simple outputs, tuck pro dials away.

Survey: Current Freight / LTL Calculators & Tools

Here are several freight / LTL calculators (or quoting tools) in active use today, along with how they work and what they reveal:

Tool / Provider	What it does / how it works	Key inputs / assumptions	Strengths (what it does well)	Weaknesses / gaps / UX pain points	Take- aways for your project
FedEx Freight Class Calculator	FedEx provides a “Freight Class Calculator” where you enter length × width × height and weight; it computes density and helps you assign a freight class. FedEx	Dimensions, weight, packaging, sometimes commodity class	Helps users estimate class based on density, reduces guesswork about class selection	Doesn’t produce full total rate; users may not see distance, fuel surcharge, accessorials explicitly	Good model: break out class/density step as a modular subtool of the full quote flow
FedEx LTL Select	Side-by-side multi-carrier LTL quotes: you submit one set of shipment info, and it returns quotes across carriers. FedEx	Origin, destination, weight, dimensions, service type	Saves on user effort (submit once), enables direct comparison	May hide details or have opaque margins; users may distrust aggregating	Use the “submit once, compare many” model, while maintaining traceability of

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FreightCenter Freight Calculator Online quote tool for LTL, FTL, intermodal; shows live rates from carriers; users input shipment specs and see comparative quotes.
[FreightCenter](#)

Weight, dimensions, origin, destination, freight class (or system estimate) may

Comparative interface; supports multiple freight modes; transparent about inputs

Accessories, surcharges or exceptions may be hidden; interface may be dense

Use their multi-mode approach for shifting between LTL/FTL in your tool; offer progressive disclosure

ODFL Rate Estimator Provides an LTL rate estimate (standard, expedited) using an

Weight, dimensions, origin & destination, tool without full

Straightforward “estimate” tool without full

Estimate only—not necessarily binding;

In your tool, clearly label “estimate” vs

	internal rate service estimator tool. speed odfl.com+1	quoting friction	may omit accessori als or special handling	binding price” to manage trust	
uShip (LTL Freight cost)	Marketplace / quoting aggregator: shipper enters shipment details, gets quotes from carriers in uShip’s network. uShip	Weight, dimensions, origin/destination, service constraints	Crowdsourced quotes, transparency of multiple offers, booking in one place	Competitive bidding can obscure how costs are built; the lowest bid may sacrifice service quality	Including carrier ratings / reliability in your quotes is useful to mitigate “lowest price trap”
TransLogistics LTL Rating Calculator	Offers quick LTL cost estimates with line haul, fuel, and accessorial considered. translogisticsinc.com	Basic shipment specs, possibly accessorial toggles	Users can see an “all-in cost” estimate	May lack full transparency of cost breakdown; potential to simplify	Good baseline for your “all-in cost” view; you can then let users dive deeper

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R+L Carriers Rate Quote	Traditional carrier quote form: fill in pickup/drop details, weight, and optional extras to get an accurate LTL quote. R+L Carriers	Weight, origin, destination, optional extras	It's authoritative (direct from carrier) and gives binding quotes	Requires many inputs and sometimes opaque in how they derive adjustments	Use "authority quotes" as a gold standard in your interface comparisons
UPS Freight / UPS Freight Quote	UPS supports freight quoting (LTL, etc.) via their site tools. UPS	Shipment specs (weight, dims, pickup/delivery)	Familiar interface, brand trust	May hide surcharges or contract-specific discounts	Offer "UPS vs others" baseline in your comparative features

ArcBest Rate Quote tool	Quoting tool for shipments (freight) via ArcBest. Arcb	Similar contexts (pickup, delivery, dims, weight)	Straightforward quoting workflow	As with many, the transparency into margin / surcharges is often lacking	Use arcs of simplicity + “explain my rate” features
CoolParcel 1 – FedEx LTL Freight Calculator (aggregate front-end)	A third-party site that wraps a FedEx LTL freight calculator (or forwards requests) and compares different routing options. Coolparcel.com	Origin, destination, dims, weight, etc.	Makes FedEx’s tool more accessible; comparison front-end	Might not always reflect real-time or contract ual discounts	Good example of “front-end aggregate or UX” — you could wrap your internal tools similarly

How These Tools Work Under the Hood (common logic & heuristics)

From the literature and publicly documented sources:

- **Density / Dimensional Weight / Freight Class**

Many tools compute *density* (weight ÷ volume) and use that to assign freight class (the NMFC class) or interpret how “space cost” contributes. FedEx’s class calculator emphasizes that. [FedEx+2FedEx+2](#)

- **Zone / Distance & Routing**

Origin and destination (zone-based pricing) is a major factor. Many carriers maintain zone tables.

- **Line Haul + Accessorials**

Basic “line haul” (the core transport cost) is then adjusted with accessorials (liftgate, inside delivery, residential surcharge, etc.). Some calculators let users toggle these extras; others implicitly assume them.

- **Fuel Surcharge / Tariffs / Market Multipliers**

Many tools factor in dynamic surcharges (fuel, congestion) on top of base cost.

- **Discounts & Negotiated Rates**

For contract customers, special discounts or negotiated rates apply. Many public calculators only show “list” or “standard” rates.

- **Minimum Charges / Marginal Steps / Class Thresholds**

There is often a floor (absolute minimum) or step changes (e.g. crossing a dimensional threshold changes pricing tier). Some tools warn users when

they cross thresholds (e.g. when a box gets slightly bigger).

- **Nonlinear scaling / bundling efficiencies**

The cost per pound or per cubic foot often decreases with scale (economies of scale), so larger shipments may enjoy favorable marginal pricing.

 Strengths & Weaknesses: What They Tend to Get Right vs Wrong

Strengths (common in good calculators):

- Fast quote generation for moderate complexity shipments
- Side-by-side comparisons (especially via aggregators)
- Good defaults (estimating class, surcharges, etc.)
- Progressive disclosure: show compact quote first, let user dig into extras
- Brand trust (carriers' own tools carry legitimacy)

Weaknesses / Gaps (frequent issues you can improve upon):

- **Opaque breakdowns** — users often see the final number but not how each component was computed.
- **Inconsistent handling of accessorials** — many extras are hidden or added later, producing surprise.
- **No “why this quote” explanation** — difficult for users to see sensitivity (which input impacted cost the most).
- **Difficulty comparing “equal apples”** — contracts, discounts, and service-level differences make two quotes not directly comparable.
- **Fixed input burden** — too many fields (commodity codes, class, handling, hazard flags) discourage casual users.

- **Poor error feedback or guardrails** — users can enter impossible dims, forget to include packaging, or mis-estimate class and get wildly wrong quotes.
 - **Static assumptions** — many calculators don't adapt to fuel price swings, regional demand spikes, or carrier capacity constraints in real time.
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🧠 Lessons & Design Heuristics for Your Tool (inspired by what works / fails)

From observing these real tools, you can derive a handful of heuristics or design guidelines to incorporate:

1. “Explain the quote” layer

For each quote, offer a breakdown: line haul, distance, class surcharge, fuel, accessorials. Also show *sensitivity* (e.g. “if weight increases 10 %, cost rises by 7 %”).

2. Threshold alerts & dimension warnings

When user dims approach a dimensional break (e.g. just over a volume threshold), show a tooltip “if you reduce width by 1 in, cost drops \$X”.

3. Default class estimation + adjustability

Automate freight class based on density and allow users to override if they have better knowledge.

4. Comparability tools / quote normalization

When comparing different carriers or quotes, normalize them to “all-in cost for same service level” so user isn't misled by hidden fees or

different terms.

5. **Progressive input flow**

Start with minimal fields (origin, destination, weight, dims), then prompt for extras only if relevant (e.g. “residential delivery?”).

6. **Error prevention & input validation**

Warn when input is nonsensical (e.g. dims too small, weight too light, missing packaging weight) before submitting.

7. **Transparency on “estimate vs binding”**

Clearly show that quotes are estimates, not final bills, especially for accessories or exceptions.

8. **Dynamic surcharges / real-time conditions**

Incorporate real-time fuel surcharge, demand-based multipliers, or zone congestion adjustments to make quotes more realistic.

9. **Trust cues & provenance**

Show carrier name, rating, service history, and optionally the margin or commission if you are an aggregator.

10. **Scenario sliders & what-if modeling**

Let users tweak variables (e.g. reduce dimensions, change service speed) and immediately see how cost adjusts.

The Current State: Where We Are

Your **needfinding** revealed three core truths:

1. **Transparency ≠ Simplicity.**

Everyone wants clarity, but not everyone wants complexity. Shippers want “why this price” in one sentence; drivers want breakdowns; corporate folks want dashboards.

2. **Users don’t want *math*, they want *fairness*.**

They’re not chasing exact numbers—they’re chasing trust. They want to feel confident they aren’t getting ripped off, even if they can’t audit the math.

3. **Decision context changes everything.**

Drivers, SMB shippers, and corporate architects use different heuristics (profit margin, time, or design impact), but they all crave insight.

Your **brainstorming** translated that into three promising tools:

- **Driver Profitability Co-Pilot** (mobile)
- **Shipper Compare & Explain** (web)
- **Ops/Design Pricing Lens** (internal analytics)

Now we refine these into a *cohesive ecosystem*, not three separate apps.

Step 1: Integrate the Core Insights Across Personas

Shared Foundation: Transparent Freight Intelligence Engine

Behind the scenes, all three tools can run on the same model—a “**freight transparency engine**” that takes shipment data, pulls real or historical carrier rates, and returns both a *total cost* and a *rationale* (“fuel cost spike, long-haul region, low backhaul density”).

This single data spine supports all front ends.

Persona	Interface	Core Question	Shared Backend Function
Driver	Co-Pilot (mobile)	“Will this load make me money?”	Profit simulation + route cost modeling
Shipper	Compare & Explain (web)	“Am I paying a fair price?”	Multi-carrier quote normalization
Business Architect / Designer	Pricing Lens (internal)	“Where do design inefficiencies waste money?”	Aggregated task-time → cost conversion

By unifying the engine, the UX stays specific but the system gains *consistency and scalability*.

Step 2: Apply Lessons from Current Freight Calculators

Based on the competitive analysis:

- **FedEx / ODFL** are good at *data entry validation* but poor at *interpretability*.
- **FreightCenter / uShip** are good at *comparisons* but opaque about *why* one price beats another.

You can outperform them by adding:

1. **“Explain My Rate” micro-feature.**

Every quote includes a natural-language summary:

“\$615 total—higher because pickup zone has low density, diesel prices +7% vs. average.”

That single sentence embodies *transparency and education* at once.

2. **Sensitivity Meter.**

A quick slider showing which variable affects cost most (weight, distance, service speed). This gives users control—and teaches them *how pricing works*.

3. **Scenario Simulation.**

In both Co-Pilot and Compare views, let users adjust inputs live—see “what if I wait one more day?” or “what if I consolidate shipments?”

4. **Progressive Disclosure.**

Keep early screens simple (origin, destination, weight). Only reveal extras (fuel surcharge, liftgate, etc.) after users opt in. That fixes the cognitive overload common in freight calculators.

5. **Trust Indicator.**

Color-coded fairness badge (green = competitive; yellow = above market; red = non-competitive). Drivers and SMB shippers interpret fairness visually, not numerically.

⚙️ Step 3: Evolve Each Prototype with These Additions

1. Driver Profitability Co-Pilot

Add:

- **Dynamic “Fair Pay” meter** (compare your offered rate vs. market average on that lane).
- **Diesel Forecast widget** (shows expected weekly fuel impact).
- **Community rate validation:** crowd-sourced lane averages (“Others earned \$1.84/mi last week”).
- **Auto-class calculator** (for mixed freight, to estimate NMFC class and density).

Result → Moves from “personal calculator” to “smart negotiation coach.”

2. Shipper Compare & Explain

Add:

- **“Explain my rate” tooltip** for every carrier’s quote (fuel, region, accessorials).
- **Interactive service slider:** adjust delivery speed to instantly see how much cost changes.
- **Packaging optimizer:** “If you reduce dimensions by 2 inches, you save \$4.50.”
- **Trust index:** aggregates carrier reliability + average delay days.

Result → Moves from “price lookup” to “decision support assistant.”

3. Ops/Design Pricing Lens

Add:

- **Data bridge to live operational rates** (use the same freight engine as above).
- **“Cost per UX error” converter:** visualize how input mistakes or duplicate work inflate logistics costs.
- **Simulation sandbox:** toggle design improvements (auto-fill, address validation) and show projected annual savings.

Result → Moves from “analytics viewer” to “financial-UX translator.”

💡 Step 4: Position the System as a Unified Transparency Suite

Your three tools together could evolve into a **platform** with a simple mission:

“Make freight pricing predictable, fair, and explainable — for everyone in the chain.”

This vision directly connects to your interview themes:

- Darrius and Dewayne’s need for **profit fairness**.
- Tina’s need for **multi-stakeholder visibility**.
- User’s need for **design-measurable KPIs**.
- Mark’s need for **simple comparisons** without noise.

That cohesion will make your final project narrative stronger: each persona gets a view of the same “truth layer,” expressed in language and visual metaphors suited to their expertise.

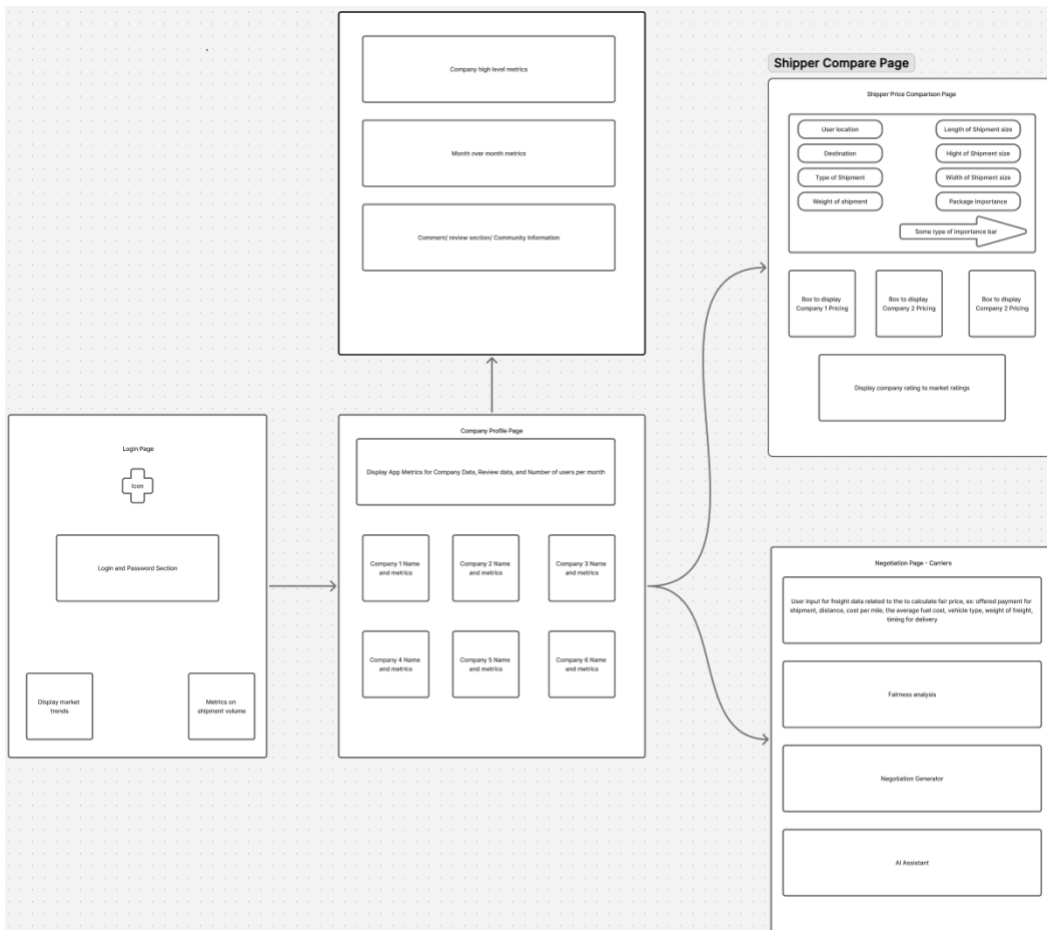
🧠 Step 5: What to Prototype Next

For your next milestone:

1. **Low-fi wireframes (3 screens each)**—show how “Explain My Rate” and “Fairness Meter” appear.
2. **Paper or clickable prototypes** for one use case: a shipper inputs a package, sees a multi-carrier comparison, tweaks inputs, and watches fairness change.
3. **Pilot testing** with 1 driver, 1 shipper, 1 FedEx employee—observe comprehension and trust levels.

4. Initial Prototyping

4.1 - Low-fi wire frame



5. Evaluation Cycle One

Main Topics Discussed

1. Review and Feedback on Shipment Price Transparency Mockups

a. Low-Fidelity Mockups Walkthrough

- Two different versions ("pages/views") of a Freight and Shipper Price Transparency page were presented for feedback, aiming to address user pain points identified in prior interviews.

b. Detailed Input Fields and Expected Behavior

- **Shipment Details Section:**
 - **Origin/Destination:** Clear and expected to be zip code entries.
 - **Shipment Type:** Stakeholder expected a dropdown with shipping speed/method (e.g., ground, air/priority).
 - **Weight/Dimensions:** Stakeholder consistently chooses package sizes below 12x12x12 inches due to drastic carrier price jumps above this cubic threshold (especially USPS, FedEx, UPS). The exact dimensions are typically known.
 - **Package Importance:** Perceived as commodity description/value, but stakeholder suggested adding a specific "value" field for insurance estimates (most carriers insure up to \$100; higher values require additional purchase).
 - **Package Type:** Indicated importance for rate calculation, i.e., mailer vs. box vs. oversized box.
 - **Insurance:** Critical for anything valued over \$100; no shipment above that value is sent uninsured.

c. Rate Comparison Features

- Stakeholder expects to compare rates for the selected shipment method among USPS, UPS, and FedEx for typical small business. For freight (>150 lbs), expectations may vary.
- **“Get Rates” Button:** Understood as fetching and comparing rates across carriers.
- **Rate Comparison Table:** Stakeholder wants clarity about which service (e.g., “ground”) is shown for each carrier.

d. Cost Breakdown and Analysis Tools

- **Price Fairness/Analysis Bar:** Understood as carrier rate vs. market average, but more relevant for freight than small parcels.
- **Show Cost Breakdown by Variable:** Expected to see shipping base rate, fuel surcharge, insurance, plus breakdowns by package weight/dimensions.
- **Dimensional-Based Pricing:** Noted as the new standard (across major carriers), with dimensions often being a bigger price driver than weight.

e. Interface and Usability Preferences

- Origin and destination entry should be side-by-side for ease of tabbing; then weight, then dimensions.
- Preference for concise dimension entries, not oversized or spread out.
- Package importance is less relevant; insurance/value is more important pragmatically.
- Suggested flexibility to choose or switch service level by carrier (e.g., click in a rate cell to reveal “ground,” “priority,” “2-day,” etc.).

f. Real-World Use Cases and Analogies

- Referenced tools like Pirate Ship, which shows consolidated carrier rates and options, influencing decision based on both cost and convenience (e.g., pickup availability).

- Examples where speed or reliability may matter more than cost (e.g., farm equipment replacement before rain).

2. Freight Community Profile and Company Directory Pages

a. Freight Community Profile Page Options

- Two versions reviewed, both offering company performance data, reviews, and reliability metrics.
- **Header Navigation:** Expected to provide access to dashboard (user/carrier activity), company profiles, and “insights” (interpreted as reviews).
- Insight input is split between customer (shippers) and carrier (service providers) reviews.

b. Company Detail Insights

- Information like “average fairness,” “reliability,” “total reviews,” and “verified reviews” considered valuable.
- Reliability should be clearly deconstructed (on-time delivery vs. goods intact/damaged).
- Regional performance data appreciated due to variable carrier/service center performance across the country.

c. Value of Extra Economic/Operational Data

- Fuel costs and surcharges are major factors in freight pricing.
- Reliability (both in time and in preventing product damage) can often outweigh price, especially for critical shipments.
- Noted that direct company contracts (e.g., with Walmart) involve unique pricing not suited to market comparison features.

d. Directory Functionality

- Directory intended as a “yellow pages” for logistics: every known carrier with the ability to “follow” companies.
- Tags/pills (e.g., temperature controlled, expedited, etc.) and coverage area are useful for filtering and discovery.

3. Freight Negotiation & Rate Calculation Page

a. Intended Audience and Function

- Designed for owner-operators, independent carriers, and those negotiating loads, not large shippers under contract.
- Fields include: origin, destination (with detailed address), calculated distance, deadhead miles, offered rate, per mile cost, vehicle type, load weight, urgency.
- **Suggested Offer and Profit Breakdown:** Tool provides suggested market rates, fairness (vs. market), and profit projections after factoring fuel and other costs.

b. AI Support / Negotiation Guidance

- Proposed inclusion of AI-generated suggestions for negotiation, rate calculation, and alternative offers.

4. UX and Design Process Observations

- Emphasis on collaboration with UX/UI professionals and recognizing their role in effective screen/interface design.
- The presenter affirmed the importance but is focused primarily on business architecture.

Detailed Recommendations and Insights

Shipment/Rate Transparency Tools

- **Key Data Fields:** Zip codes for origin/destination, package type, accurate dimensions under the 12x12x12 threshold, and explicit package value.
- **Default shipping option:** Should likely default to “Ground” for small parcels with ability to explore upgrades if users wish.
- **Rate Explanation:** Need to clarify why rates might be unexpectedly high (e.g., oversize/dimensional pricing).
- **Service Offerings:** Allow users

to swap between different service/products by each carrier easily to see rate impacts.

- **Subjectivity of “Importance”:** Field could be dropped/replaced with explicit “Value” (for insurance).
- **Segmentation:** Consider splitting interfaces for small parcel vs. freight/bulk shipments due to very different user needs and priorities.

Company Profiles/Community Pages

- **Performance Metrics:**
 - Reliability should be granular (on-time vs. undamaged delivery).
 - Regional views provide genuine value for shippers with area-specific needs.
- **Review Sourcing:** Explore integration of aggregated public reviews (Google, Yelp) for broad “market” perspective.
- **Peer Comparison:** Show user’s most-used/followed carriers for targeted benchmarking.
- **Profile/Directory Navigation:** Users can browse the universal directory, but should also have a dashboard for followed/preferred carriers.

Freight Negotiation

- **Applicability:** Most relevant to “open market” or non-contracted freight operators.

- **Key Variables:** Integrated calculation of fair market rate, actual costs, and expected profit after deductions.
- **Negotiation Support:** AI-driven suggestions and “counter-offer” helpers are innovative features.

Action Items

1. **Add “Value” field** in shipment details for calculation of insurance needs and cost.
2. **Refine tab/field ordering:** Place origin and destination side-by-side, then tab to weight and dimensions.
3. **Allow easy toggling or selection** of service/product (ground, priority, air) per carrier when viewing rate comparisons.
4. **Revise/replace “package importance” bar** with an explicit insurance/value field and options.
5. **Enhance “Show Cost Breakdown”** to detail both carrier-specific surcharges (fuel, insurance) and variable-driven impacts (e.g., exceeding dimensional thresholds).
6. **Develop user “dashboard” page** to show followed companies and dynamic benchmarking.
7. ****Explore integrating third-party market/review**

data** (e.g., Yelp, Google) for broader market insight. 8. **Segment customer flows** for small parcel vs. freight to ensure relevant information and tools display. 9. **Split “reliability” metric** into “on-time” and “non-damaged/OS&D” for clearer operational comparison. 10. **Iterate on directory/profile navigation** for clarity between universal listing and personal portfolio/dashboards.

Follow-Up and Additional Points

- Potential further integration with UX/UI experts, especially for expanded user testing and feedback loops.
- Additional iterations planned for deeper dive into freight negotiation and rate transparency, possibly with more advanced AI-guided components.

- No specific follow-up meeting scheduled but possibility discussed for connecting with company UX team or exploring more architecture-focused features in the future.

Miscellaneous

- Discussions included anecdotes from stakeholder's shipping experiences, business practices, and personal insights from logistics industry work with FedEx Freight.
- Mockups were produced and reviewed in Figma, which is also used internally at FedEx.

PART 1. Interviewer questions (in order, grouped by topic)

Freight & Shipper Price Transparency page (Page 1)

1. "Within the shipment details... what does this first kind of like text box mean to you? The origin. Origin zip code?"
2. "And then the destination would be...?"
3. "And then the shipment type — if that was a drop down, what would you expect to see?"
4. "And then from this side of it you have what is like this weight indicator kind of mean to you?"
5. "And then yeah, the dimensions here. Is this something that you would kind of know offhand?"
6. "We have the package importance. How would you use this... bar? And what does this mean to you?"
7. "Within this box, is there anything that you might want to see that you feel like is missing, or is there a specific box that isn't clear... or do you think this is the right amount?"
8. "And then we have this button here, 'get rates.' What does that mean?"
9. "For the rate comparison... is there any information that you would want to see to help with the decision process or anything you think might be worth adding... to help with the comparison of carriers?"

10. "Lastly on this page we have kind of like the price fairness analysis. What does this bar mean to you or what do you think it represents, from your perspective?"
11. "Is that what it's intended for?"
12. "If you were to click 'show cost breakdown by variable,' what would you want or expect to see?"
13. "Does this make sense to you or should it be worded differently?"
14. "Would you expect to see those rate changes in the rate comparison based upon the dimensions or the changes that you have here?"

Freight & Shipper Price Transparency page (Page 2 / alternate layout)

1. "When we look at the second option... all the boxes are pretty much the same, but the function here is different. Would you want to at least provide me your thoughts on what this means to you and how this bar will function?"
2. "So would you expect like an extra box for the type of shipment that you're expecting?"
3. "Below we have the rate comparison... what are your thoughts on this specific tab? Is there any information from the other screen that should be here, or if you hit that dropdown what would you expect to see or visualize?"
4. "And then the price fairness analysis tab here — what are your thoughts on this tab?"
5. "So between the two pages of the freight and shipper price transparency, which one do you like more? Or are there components within one that you like more than the other?"
6. "Gotcha... [later follow-up]: Would you best identify it as [product or service]?" (This was you trying to label "service type vs product type," but it's still a clarifying question about terminology.)

Freight Community / Company Profile page (the review / network / dashboard concept)

1. "When we look at the very top... what would you expect to kind of see here and what do these different buttons or links mean to you?"
2. "We have kind of like the company profile page here. What do you like about this? What information would you like to see? Is this enough information? What are your thoughts on this?"
3. "Am I off base there?" (You asked this when you explained Dashboard / Companies / Insights.)
4. "So... yeah... So basically you're getting insights based on the app. So is that...?" (You're checking if the company profile layout matches her mental model. Functionally also a question.)
5. "So... what's 'carrier' mean in the insights work there?"
6. "Yeah... so for this aspect of it [the economic info/fuel index], are there other economic data you would like to see for more transparency?"
7. "Anything else for this page?"

Freight Community Profile page (Option 2 / alternate layout)

1. "For the second option... this is pretty much the same thing, but just a different example. [User comments on regional performance.] ... On this other one where you just have the one [graph], are you able to toggle and get different information or is that just static?"

(You're basically asking her expectation of interaction/toggling.)
2. "Market and peer comparison — is that going to generate based off of your companies that you follow? ... What would you expect?"
3. "Anything else between these two pages before I go to the last one?"
4. "So company directory — in your opinion are these the companies you follow and how to contact them?" / "Or is this every company Google can find...?"

5. “Quick question: if you scroll back up to the other page and you hit ‘follow company,’ where are the companies that you follow?”
6. “So... is the dashboard then the dashboard of the companies you follow?”

Freight Negotiation page (the broker/quote/offer calculator)

1. “We could jump back to [the negotiation page]... I want your customer mindset. So... this kind of talks about the freight negotiation pages... So when you look at this: you put your origin, destination... [You ask:] This probably needs to be zip codes?”
2. You explain the screen, then: “So people can go in here and compete for shipments?”
3. “If you are meeting with a broker and they offer you \$1500 to take Dallas → Corpus Christi... you can input the information... [You walk through it and then...] So this would give you the suggested offer — the suggested offer that you should be charging people, right?”
4. “Then here... fair pay analysis... [Is] down here... market trend, fuel cost, insurance cost... profit breakdown... Then negotiation tactics... [implicit question: does this make sense / would you use this?]”
5. “What if you’re shipping with a company that you already have a pricing contract with — then [is this still for you]?”
6. “Counter-offer generated... okay... [implied] is this useful?”

That’s what you asked. You hit coverage in these clusters:

- What do you think this field means?
- Would you use this, and how?
- What’s missing?
- Which layout/version works better, and why?

- What would you expect this button/tab/dropdown to do?
- Is this relevant for how you actually run your business?

That's textbook formative evaluation / concept validation.

—

PART 2. What was discussed (thematically summarized with direct implications)

I'll break this into four design areas you walked through:

- A. Shipment pricing / rate comparison screen
- B. Cost breakdown & fairness analysis
- C. Community / carrier profile network
- D. Negotiation & fair pay tool for carriers / brokers

A. Shipment pricing / rate comparison screen

Working theory of user behavior here: The small business owner is shipping parcels, not freight pallets, so their brain model is "I ship an item to a buyer, I just want the cheapest option that won't get me yelled at."

Key takeaways:

1. Inputs make sense, mostly.
 - Origin zip = "my zip." Destination = where I'm sending it.
 - Weight and dimensions = final packed box. She absolutely knows her standard box sizes and mailers.
 - She cares a lot about dimensions because carriers now use "dimensional weight pricing" (billing based on volume, not just pounds).

She specifically optimizes all her packaging under 12 x 12 x 12 inches because going over that jumps cost fast.

Design implication: “Dimensions” is not optional fluff. It’s critical, and must be easy to type.

2. “Shipment type” needs clarity.

- She first thought “shipment type” meant speed (Ground vs 2-Day vs Overnight).

- Later she saw it could mean “package type” (box vs poly mailer).

- She said both actually matter: service speed changes price a lot, and packaging type affects eligibility for certain rates.

Design implication: You actually have two concepts:

(a) service level / speed (ground, 3-day air, overnight)

(b) physical container / package class (padded mailer, standard box, oversized box)

Those should not live behind one mysterious dropdown called “shipment type.” Split them or label clearly.

3. “Package importance” is not how she thinks.

- You tried to capture “how critical is this shipment / how replaceable is it.”

- She translated that to “is this thing rare vs replaceable.”

- Then she basically killed it: her customers don't talk like that. For her scale, what matters is declared value and extra insurance.

- She wants: "What's the value of the shipment?" + "Do you want to add extra insurance beyond the default \$100 coverage?"

Design implication: Replace "package importance" with "Declared value / add insurance?" This ties directly to cost and risk.

4. Workflow preference.

- She likes origin and destination next to each other so she can tab through quickly.

- Then weight.

- Then dimensions.

Basically: Location → size/weight → service. She wants to data-entry this fast off a scrap of paper, not hunt around the UI.

Design implication: Input layout matters for muscle memory. You can score real usability points here.

5. Comparison expectations.

- Her normal shopping behavior is literally: USPS Ground Advantage vs UPS Ground vs FedEx Ground Economy.

- She expects to see each carrier's named service SKU (not just carrier name), plus price.

- She also sometimes chooses a slightly more expensive carrier simply because of convenience (ex: USPS picks up at her house; UPS requires her to drive somewhere).

- Secondary need: show options for faster service (2-day, overnight) so she can upsell or answer “can you get it here in two days?”

Design implication: The comparison table should:

- Show carrier + specific service level.

- Default to cheapest ground options.

- Let her toggle/expand to “faster but more expensive” offers.

- (Bonus) Add a “pickup / dropoff convenience” note. Convenience beat price by one penny in her real story.

B. Cost breakdown & fairness analysis

This is where your “Explain my rate” and “Price fairness analysis” ideas live.

Her reactions:

1. “Explain my rate” is valuable.

- She wants to see base rate, fuel surcharge, insurance, dimensional upcharge, etc.

- She especially wants to know, “Why is this one so expensive?” so she can fix it by using a smaller box.

- She literally did dimensional math in the interview. She is actively managing cubic inches to control cost.

Design implication: “Explain my rate” should:

- Call out what’s driving cost.
- Highlight when you’ve crossed a size threshold (“This box exceeds 1728 cubic inches, so you’re paying an oversize surcharge.”).
- Suggest actionable next steps (“Use a smaller box and save \$X”).

That’s gold. That’s retention.

2. Live tinkering.

- She expects that after clicking “show cost breakdown by variable,” she can tweak dimensions/weight and immediately see new rates.
- She’s basically asking for a lightweight what-if simulator.

Design implication: That modal/screen shouldn’t just be static text. It should be interactive recalculation.

3. “Price fairness analysis” is situational.

- For her parcel business (eBay-style sales of normal goods), she does not care about “is this a fair price vs the market.”
- Her buyers just want “cheapest that shows up reasonably intact.”
- She thinks fairness / breakdown will matter much more for heavy

freight, like shipping a 200 lb engine or palletized freight, where cost is in the hundreds or thousands and delays have real business impact.

Design implication: You're actually designing for two different shipping modes:

- Parcel/e-commerce: optimize for cheapest ground, pickup convenience, and insurance.

- Freight/industrial: optimize for transparency, fairness, liability, and timing guarantees.

You probably need mode selection early ("Small parcel <150 lb" vs "Freight / pallet / engine / heavy equipment"). She basically told you to fork the UX.

C. Community / carrier profile network

This is your "Freight Transparency Network / Company Profile / Reviews / Dashboard / Directory" concept.

Here's how she understood it (which is how you know your information architecture is mostly working):

1. Header navigation

- "Dashboard": she thinks this would be her personal view — which carriers she uses, how much freight she's giving them, reliability, etc.

- "Companies": the carriers themselves / list of trucking companies.

- "Insights": she interpreted as performance data and reviews, and she's comfortable with the idea of public reviews.

Design implication: You're on the right track with those labels. Just define them clearly in onboarding so users form the same mental map she did.

2. Company profile card/page

- She likes: average fairness, reliability, total reviews, verified status.
- She explicitly said verified matters because it prevents random fake companies.
- She expects “Follow company” to add that carrier to her personal dashboard for quick monitoring later.
- She likes being able to “Add your own insight / review,” and she understands the two perspectives:
 - As a shipper (customer buying transport)
 - As a carrier (doing hauling for someone else / subcontract work)

Design implication: You surfaced a legit two-sided market:

- Shippers reviewing carriers.
- Carriers reviewing shippers/clients.

That's actually unusual and powerful because it builds accountability in both directions.

3. Fairness / reliability metrics

- She asked you to split “reliability” into two sub-metrics:
 - On-time / speed (did it arrive on time)
 - Condition / damage (did it arrive intact)
- In freight world this is literally tracked under OS&D (“over, short, and damaged”), which is a real internal FedEx KPI for pallets.

Design implication: Reliability should not be a single score. Make it at least:

- On-time %
- Damage rate %

Users who ship engines and pallets care about both separately.

4. Regional performance

- She loved the “regional performance” view in version 2 of the company profile.
- She said some service centers/regions perform worse (slow, more damage, more expensive behavior, etc.).
- That means: a carrier might be “great in Midwest, sketchy in Northeast.”

Design implication: Regional reliability maps are a serious differentiator.

She's telling you that ops quality is not uniform nationwide, and shippers actually know this from experience. You can expose it.

5. Market and peer comparison

- She expects “Market” = all known public options / scraped review sources (Google, Yelp, etc.).

- She expects “Peer comparison” = just the carriers she actually uses or follows.

- She wants to see her top few carriers head-to-head.

Design implication: That lower-right “Market vs Peer” widget is valuable but needs to be clearly scoped:

- “Here's how YOUR chosen carriers stack up.”

- “Here's how the general market looks.”

6. Directory vs Dashboard

- She thinks the big company directory is like “the yellow pages of trucking,” which is correct for your intent.

- She then immediately looked for “my followed companies” and assumed that would live in Dashboard.

- She expects tags like “temp controlled,” “expedited,” coverage area, etc. She explicitly liked the little “pills.”

Design implication: You're very close here. You just haven't drawn the

personal dashboard screen yet, but the mental model is already there in her head. That's good pressure on you to make that page.

D. Negotiation / "fair pay" page for carriers (the broker/offer calculator)

Working theory: This screen is not for eBay sellers. It's for people who haul freight and negotiate per-load rates.

How she reacted:

1. She understood it fast once you walked it.
 - Input: origin, destination, deadhead distance, load weight, vehicle type, urgency.
 - Broker offers you, for example, \$1500.
 - The tool calculates cost factors like fuel, distance, etc. and tells you "suggested offer," i.e. what you should counter with.

That matches how actual load board / spot market conversations work.

2. She framed it as "competing for shipments."
 - Her first instinct was: does this let drivers bid?
 - Your clarification was: it's more like "Are they lowballing you? Here's data you can use to push back."

Design implication: The screen is basically a negotiation coach for independent carriers / small fleets. That's a strong pitch: "Know your worth before you say yes."

3. She said: if you're already in a locked contract with a large shipper, you're not negotiating load-by-load like this.

- So, yes, this tool is for non-contract, spot freight, gig-style hauls.

Design implication: You might want an up-front selector like:

- “Contract / agreed lane rate” (then you just track margin)

- “Spot / brokered load” (then you open negotiation assistant)

4. She liked the “fair pay analysis,” “market trend,” “profit breakdown,” and auto-generated counteroffer.

- That's exactly what an underpowered independent driver needs when a broker tries to push them down.

- This echoes the same theme from earlier: transparency and leverage for the little guy.

Design implication: You are basically designing a labor tool as much as a logistics tool. That is ethically interesting (you're helping them not get underpaid), and also commercially very sticky.

Attendees

- **Alex:** Presenter/designer (driving conversation, asking for feedback)
- **Chai:** Primary reviewer/user, giving feedback

Main Topics Discussed

1. Freight Rate Comparison and Shipment Details UI

a. Shipping Details Section

- **Fields Discussed:** Origin zip code, destination zip code, shipment type, weight, dimensions, package importance.
- **Feedback:**
 - Zip code entry is sufficient but allowing address input could be helpful for users not knowing the zip.
 - Service type missing from the dropdown, initial expectation was overnight/standard/economy options.

Distinction needed between shipment type (envelope/parcel, special items) and service speed.

- Package importance (slider): Interpretation was unclear; could relate to value (for insurance) or timeliness.

Requested clarity via an info icon; sliders should be tied to specific services (e.g., tracking, insurance).

Suggested bundling services for each importance option or allowing custom selection.

b. Rate Comparison Table

- Service speed types not fully represented; overnight missing, potentially misleading.
- Importance of clear rating origins (star ratings) and tying them to actual user experience data, not just aggregate sources like Google/Yelp.
- Ability to view full written reviews would be beneficial, especially to assess issues like timeliness or damage.

- User wants to filter/see only relevant services (not 3–5 day if they're seeking overnight).
- **Cost Breakdown:** Expectation is a line-item breakdown (base rate, insurance, extras).
- Question about multi-package rates: Is there an advantage or bulk discount for multiple packages versus a larger consolidated box?

c. Price Fairness Analysis

- Users favor seeing where their rate stands versus market (below average/fair/above), but the color scheme should be intuitive (green for fair/below, yellow for mid, red for above).
- Query about

how market average is calculated (daily rates, seasonal effects, etc.).

- User expects a dynamic comparison (not just historical average, but in-context with current rates), especially considering holidays/peak periods.

2. Alternate Rate Comparison UI Options

- **Preference for Option 2:** Cleaner design, better use of logos, more visually appealing.
- Color inconsistency across fairness bars can cause confusion; ensure alignment.
- If more than three rate options are presented, concern about screen real estate and cognitive load with too many fairness bars or variable breakdowns.

3. Freight Community Profile and Ratings

a. Company Profile Page

- Clarified that the profile is for logistics companies, not individual users.
- Ratings for:

- Price fairness—clarified it's 1–5 stars, above average is 5.
- National average meaning (needs clearer terminology).
- Reliability and "verified" reviews (user-origin).
- Suggestion to reconsider label conventions, e.g., “price fairness” instead of "average fairness".

b. Performance Dashboard

- **Trend graphs** need timeframe (week/month/year) for context.
- On-time payment, communication, broker average: all make sense if scored 1–5.
- **Regional performance colors/legend:** Some confusion on where legend is referenced and how it relates to dashboard vs. community insights.
 - Suggestion: move legend to community insights or add toggle functionality for region-based drilldown.

c. Feedback/Review Sharing

- Default is anonymous sharing, preference to keep this as default.
- Clarified regional tagging is by shipment origin for community feedback; discussion on pros/cons of origin vs. destination-based insights.
- Consistency of filtering by region for relevant insights deemed important.

d. Market Peer Comparison

- Clarified companies listed for peer review should be clickable.
- Fuel index inclusion seen as valuable for carriers due to regional fuel cost variances.

*Alternative Community Profile Layout**

- Preference for cleaner layout showing month-over-month trends and a clearer performance dashboard.
- Verification status, anonymous/repeat partner badges explained; UI cues (e.g., clickability of company names) need to be clear and consistent.

4. Dashboard & Filtering Functionality

- Dashboard allows searching/filtering for companies (by type, region, service area, etc.).
- Feedback that knowing the service area (not just HQ location) is most useful for real-life carrier selection.
- Request for filters based on all the "bubble" attributes visible (e.g., LTL, intermodal, etc.), with service area as top priority.
- Acknowledged that inconsistent company self-presentation may hinder side-by-side comparison, but personalized highlights are a positive.

5. Carrier Profitability/Negotiation Tool

- Quick demo/review of load negotiation calculator for carriers (offer rate, per-mile calculations, fuel price, profit breakdown).
- Highlighted "conservative," "recommended," and "aggressive" approaches/tools for negotiation.
- Very positive feedback: sees immediate, practical benefit – "favorite part" of the platform.

Detailed Feedback and Insights

Shipment Details

- Clarification on input: allow address-to-zip flexibility if possible.
- Add missing service speed selector and/or clarify with user-facing help features.
- Package importance: strongly correlate each option with visible, specific service upgrades (e.g., insurance, GPS tracking, notification frequency).

Rate Comparison

- Must clarify data sources and timeframes for rates and reviews.
- Allow granularity in cost breakdown; show base cost vs. insurance/other surcharges.

- Enable filtering of displayed service types.
- Consider multi-package shipment calculation features, if feasible.

Price Fairness & Trends

- Color-coding must be consistent and logical.
- Show not only intra-app comparison but also recent historical market context.
- Show explicit timeframes in trend charts for actionable decision making.

Community/Peer Review

- Badging (verified/anonymous/repeat) should be clear to users.
- Clarify national average and benchmarking logic.
- Regional data (performance, insights) should always be surfaced by service area (origin), with option for destination if possible.

Dashboard and Filters

- Comprehensive filtering is critical: region, carrier type, service, and custom attributes.
- Balancing custom company highlights with data consistency is important.

Carrier/Negotiation Tools

- Feature is standout; visually clean, actionable insights and scenario planning for negotiations highly valued.

Action Items

1. UI Improvements

- Ensure option 2 of rate comparison and community profile designs are primary, as user preferred for cleanliness and clarity.
- Unify color coding for fairness analysis; avoid confusion between green/yellow/red indicators.

- Add timeframes to trend charts and dashboards.
- Make company names in peer comparison clearly clickable (button-like appearance).

2. **Feature Enhancements**

- Add service speed selection to shipment details.
- Clarify slider/importance features with info icons and clear link to bundled services.
- Enable address-based entry for shipping, with auto-zip lookup if needed.

3. **Filtering/Comparison**

- Expand dashboard filtering options: region (by service area), type, company attributes.
- Ensure filter logic is user-friendly and prioritized by user needs.
- Consider supporting multi-package rate comparisons for advanced users.

4. **Community/Insights**

- Clarify regional insights logic (use origin, but investigate destination as optional).
- Use more intuitive legends and UI grouping for region-specific data.

5. **Information & Data Sources**

- Be transparent about source and calculation period for market rate comparisons, price fairness analysis, and reviews.

Follow-Up Items

- Consider collecting further feedback from users with more logistics expertise, especially around the community/comparison features.
- Share design updates reflecting this feedback with Chai and/or wider user test group.
- Investigate technical feasibility of integrating live/address-based zip code detection and multi-package pricing.

Shipment Details & Rate Comparison

1. “When you look at the origin zip code, what does that mean to you?”
2. “And the destination zip code — what would that mean to you?”
3. “Would you have any recommendations to these fields? Is there anything you’d like to see?”
4. “Would you rather see the actual address or is the zip code good enough?”
5. “When I have shipment type here, what does this mean? If I was to select this dropdown, what would you expect to see?”
6. “Would you expect to see service types or parcel types?”
7. “Based on that, is service type something missing that would provide value?”
8. “Is there any other feedback related to the other tabs within the space?”
9. “When you look at this ‘package importance,’ how would you expect this to function? What does it mean to you?”
10. “How would you expect this bar or slider to function? How would you use it?”
11. “Is there a better rating system that would be ideal to replace this?”
12. “What would be ideal here to help paint that picture if you were a customer or user?”
13. “Would the information icon or a toggle make that clearer?”
14. “Anything else you want to add for the shipment details on the first page?”
15. “What are your thoughts on this rate comparison section of the web app?”
16. “Reviews from where — by who?”
17. “What are your thoughts on this price fairness analysis bar? What does it mean to you?”
18. “Do you need to pick one carrier before it gives you the price fairness analysis?”

19. "So, are you pulling the market average on a daily basis or seasonally?"
 20. "If you select 'cost breakdown by variable,' what would that mean to you and what would you expect to see?"
 21. "Anything else you want to add to this page?"
 22. "Are there any price differences for sending multiple packages vs just one?"
-

Layout Comparison (Option 1 vs Option 2)

1. "What are your thoughts when you look at this layout compared to the other one?"
 2. "Which one do you like more?"
 3. "Why do you prefer that one?"
 4. "Anything else you'd like to see within this space?"
 5. "What are your thoughts on this price fairness analysis bar compared to the other one?"
 6. "Would a different color scheme make more sense?"
 7. "If there are many rates, would showing all at once feel cluttered?"
 8. "If you do the cost breakdown by variable, how many variables are you looking at?"
 9. "Anything we might have missed or haven't covered on this page?"
-

Freight Community Profile Page

1. "When you look at the top, what does this mean to you?"
2. "Does this feel like a company profile or a user profile?"
3. "So average fairness — what does that mean to you?"
4. "Would 'price fairness' as a naming convention make more sense?"
5. "Is national average intuitive or confusing?"
6. "Anything else you want to add here?"

7. "What does this performance dashboard mean to you?"
 8. "Would it make sense to break this out by shipper vs carrier?"
 9. "Would having time frames make trends clearer?"
 10. "How do you interpret the regional performance colors?"
 11. "Should the legend move elsewhere or toggle?"
 12. "What does this section (right-hand side, feedback) mean to you?"
 13. "Would you prefer the option to share non-anonymously?"
 14. "Would you rather insights be based on origin or destination?"
 15. "From your perspective, which is better — origin or destination?"
 16. "Anything else before moving to market peer comparison?"
 17. "What populates the fuel index?"
 18. "Would you want to see regional or national data?"
 19. "Anything else missing for this page?"
-

Freight Community Page (Option 2)

1. "What are your thoughts on this version compared to the other?"
 2. "Would month-over-month trends be more meaningful?"
 3. "How do you interpret verified, anonymous, and repeat partner?"
 4. "What are your thoughts on this market comparison section?"
 5. "Should the carrier names (e.g., C.H. Robinson, Coyote) be clickable?"
 6. "Anything missing or unclear between these two versions?"
-

Dashboard Page

1. "What are your thoughts when you look at this dashboard?"
2. "Can you filter by region or carrier type?"
3. "What would you expect when you hit that dropdown?"
4. "What kind of filters would be most useful?"
5. "What's missing here?"

6. "What would make this comparison easier?"
 7. "Would you expect to see service area or home base?"
 8. "Anything else you'd like to change or add?"
-

Freight Negotiation Page

1. "As a carrier, how do you interpret this screen?"
2. "Would you use this to negotiate offers with brokers?"
3. "Does showing conservative / recommended / aggressive help decision-making?"
4. "Which layout feels cleaner?"
5. "Any additional features or clarity needed here?"

Meeting Summary: UI Mockups & Feature Review Session

Date: October 27, 2025 **Purpose:** Review UI mockups based on user interview feedback, prioritize features, discuss usability and improvements, and select favored design options. **Attendees:** (Implied: Project Lead/Facilitator, Interviewee/end-user tester)

Main Topics Discussed

1. Overview and Process

- The facilitator summarized the process: a comprehensive review of prior user interview feedback was conducted. UI mockups were created based on this, intending to cover, as completely as possible, the discussed needs.
- The session's primary goal was to focus closely on three main UI variations/options, with extended review if time permitted.

2. Shipment Rate Comparison Interface

a. Understanding Input Fields

- **Origin & Destination Zip Code:** Clear expectation these reflect shipment starting point and intended delivery address, not necessarily the user's own location.
- **Shipment Type Dropdown:** Initial confusion about whether this covers delivery speed (e.g., next day) or special handling (e.g., refrigeration). Clarified it's for types like parcels versus skids/envelopes; noted possible ambiguity for the user.
- **Weight & Dimensions:** Usual confusion over length/width/height order and terminology; noted the need for clear labeling or tooltips. User expects cumulative weight for multi-package shipments.
- **Package Importance Bar:** User infers a left-right scale (possibly a slider) that combines urgency, need for tracking, and how irreplaceable the package is. There's ambiguity if it fully covers speed, special care, or just insurance level.

b. Key Insights and Missing Elements

- **Ambiguity in Shipment Type:** Unsure if it covers a full array of types (e.g., bulk/skid, multiple items to different locations). Suggests dynamic hints or hover contextual details.
- **Separating Speed and Importance:** User advises explicit speed (delivery time) controls; c

urrent “importance” concept feels overloaded and might conflate urgency with irreplaceability.

- **Relation to Rate Calculation:** Unclear how sliding importance directly affects rate; user suggests showing price ranges or indicative rates as importance increases for better understanding and cost sensitivity.
- **Insurance & Protection:** Wants clear information on which service level includes package protection, at what value, and whether it's included by default or at extra cost.

- **Delivery Timing Guarantee vs. Average:** Needs distinction between “delivered by” (promised date) and “on average” delivery duration. Would appreciate both, to better weigh guaranteed speed vs. cost.

c. Rate Comparison & Analysis Section

- **Breaking Down Costs:** Appreciates detailed breakdowns (e.g., base rate, fuel surcharge, insurance) but is skeptical/confused about what counts as “additional fees.” Requests transparency, especially for non-standard surcharges.
- **Price Fairness Analysis:** Recognizes value in showing if rate is fair compared to recent market history (last 30 days); visual indicators (e.g., colored dot or bar) are sufficient for most users.
- **Variable Cost Breakdowns:** Expects breakdown to dynamically reflect chosen options, such as refrigeration, tracking, or required signatures, for each rate.
- **Comparisons by Carrier/Market:** Likes feature to compare any selected carrier to the broader market for insight into competitiveness.

d. UI Option Preference – Rate Comparison Page

- Strongest preference for options grouping logical/matching fields together (e.g., zip codes side-by-side, weight and dimensions together).
- Favors cleaner, less busy layouts:
 - **Option 1:** Preferred for logical grouping and less visual clutter, especially at the bottom of the screen.
 - **Option 2/3:** Appreciates their middle content, but finds extra detail (especially at the bottom) overwhelming unless specifically sought.
- Would appreciate consolidated detail “on demand” rather than by default, and notes that minimizing required clicks for core flows is also desirable.

3. Freight Community Profile Page

a. Metrics and Trust Indicators

- **Core Metrics:** Likes split between fairness, reliability, review count, and verification status.
- Values high review counts and some indicator of authenticity (verified/unverified).
- Suggests definition tooltips, especially where terms are unusual (e.g., “fairness”).

b. Performance Dashboards

- **Fairness Trend:** Seen as useful only if it reflects value to shippers/end-users; concern about ambiguity if metrics drift due to market-wide changes.
- **On-Time Payment:** Relevance varies by user role; more valuable for contractors/carriers than shippers.
- **Reliability Score:** Considered vital for shippers—shows consistency of delivery and undamaged service.
- **Broker Average & Communication:** Valuable for B2B users or large accounts (e.g., Nike), less so for individual shippers.
- Suggests context-aware dashboards or toggles, adjusting metrics shown based on user role (shipper, carrier, broker).

c. Community Insights and Feedback Submission

- Likes ability to provide contextual feedback (role, shipment detail, comments).
- Notes 100-character limits may be restricting, especially for negative experiences.
- Some wording unclear (“I participated in this shipment”); recommends role-specific wording.
- Trending keywords valued as a summary of hot topics/issues.
- Suggests consideration of metrics such as renewable fuel/eco-friendly practices, especially as decarbonization becomes a differentiator.

d. UI Option Preference – Community Profile Page

- **Option 2/3:** Finds these cleaner and easier to digest; especially likes the “A/B” style letter grading in Option 3 for quick scanning.
- Wants key quantitative metrics up top (review counts, summaries) and clear visual status for ratings.
- Prefers option 3 overall for its immediate clarity and simplicity.

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4. Freight Community Directory

- **Company Directory:** Wants full, uncensored listings (not just “curated” or sponsored providers).
- **Key Metrics:** Likes display of high-level metrics (fleet size, monthly volume, coverage), tailored to the company’s strengths.
- Skeptical of payment-days metrics unless acting as a third-party driver; considers reliability/volume data more relevant as shipper.
- **UI Option Preference:** First option favored for its broad, diverse, and user-relevant metric set over payment-cycle-focused alternatives.

5. Freight Negotiation Assistant Tool (for Carriers)

- **Target Audience:** Tool is squarely aimed at contract drivers/carriers, assisting negotiation with brokers/shippers.
- **Features:** Displays all key route and offer parameters: route info, current rates, average fuel costs per mile, timing requirements, vehicle type.
- **AI Counteroffer Recommendation:** Highly valued; helps less-experienced carriers know when and how to successfully negotiate rates, similar to coupon code “success rates” on e-commerce.
- **Clarity Gained:** The mixed-user base (shippers vs. carriers) explains presence of “on time payment” and other carrier-focused metrics elsewhere in the app.

Additional Insights and User Suggestions

- **Mobile Accessibility:** User inquires if a mobile app is envisioned. Facilitator confirms web-first, but recognizes mobile is important for broad market reach.
- **Free vs. Paid Features:** Potential for “freemium” model (core features free, advanced analytics or negotiation tools, etc. behind paywall).
- **Sustainability Data:** Suggests including explicit green metrics (renewable fleet percentage, carbon footprint) as future interest grows.

Action Items

1. Clarify Labeling & Definitions:

- Add tooltips or info popovers for ambiguous terms (e.g., “package importance,” “fairness”).
- Make clear distinction between “average” vs. “guaranteed” delivery timing.

2. UI Enhancements:

- Combine best aspects of favored UI options (grouping, minimalism, clarity); consider “on-demand detail” for advanced metrics.
- Group logically associated input fields in shipment forms (e.g., zip codes, weight/dimensions).

1. User Role Differentiation:

- Tailor dashboard metrics based on selected role (shipper, carrier, broker); consider a toggle for perspective shifts.
- Adjust performance dashboard to make “on-time payment” and similar metrics more prominent for carrier users.

2. Feedback Section Improvements:

- Wording review for action buttons/checkboxes (e.g., “I participated...”).
- Consider increasing comment character limit or providing a multi-part response for richer detail.

3. Sustainability Metrics Exploration:

- Investigate feasibility of displaying renewable energy usage/carbon footprint alongside financial metrics.
- 4. **Rate Breakdown Visibility:**
 - Ensure clear cost breakdowns available for all carriers; clarify unusual or one-off surcharges; possibly via an expandable info section.

Follow-Up Points

- **Future Session:** If time allows, review additional three mockup options that were not covered today.
- **Sustainability Metrics:** Schedule brainstorming/requirements gathering for adding environmental metrics.
- **Mobile Strategy:** Assess tools and team capacity for parallel mobile app development.

Decisions Made

- **UI Preferences:**
 - Rate Comparison: Strongest preference given to option 1, with minimal, clear grouping.
 - Community Profile: Option 3 favored for visual clarity and simple grading.
 - Company Directory: Option 1 selected for diverse, relevant metrics.
- **Functionality:** AI negotiation aids, cost breakdowns, and fairness analysis seen as highly valuable for at least some user segments; should be preserved or enhanced.

User S Feedback

Recorded on: Thursday, October 30, 2025

Meeting Summary

Date: October 30, 2025 Topic: Freight and Shipper App Wireframes – Design Review & Feedback Session

Attendees

- Designer/Project Lead (Primary speaker)
- Peer Reviewer (Feedback provider)

Main Topics Discussed

1. Overview and Purpose

- Designer presented "extremely low-fi" wireframes for a freight and shipper app.
- The designs include: a login page, company profile page, dashboard, shipper comparison page, and freight negotiation page.
- Aim: Deliver more than the basic project requirements to ensure a top grade, requesting detailed peer feedback.

2. Walkthrough of Shipper Price Transparency Page

Key Features & User Flow

- **Shipment Details:**
 - Origin Zip Code
 - Destination Zip Code
 - Shipment Type (drop-down): Expected options—overnight, two-day, etc.
 - Weight & Dimensions inputs
 - Package Importance (slider/bar—functionality unclear initially to reviewer)
- **Get Rates Button:** Submits shipment details to obtain rate comparisons.
- **Rate Comparison Section:**

- Lists shippers, service types, and prices.
- "Explain My Rate" drop-downs detail breakdowns (base, fuel, insurance).
- "Protected" tag indicates insurance included.
- **Price Fairness Analysis:**
 - Visual comparison (bar/graph)—compares individual shipper rates to market average.
 - Discussion if it should be per-shipper and more instantly visible (e.g., % difference on each card).
 - Reviewer suggested more interactive features and visible comparisons.
 - Functionality to select or add other shippers for comparison suggested.
 - Possibility of expandable/collapsible cost breakdowns and added market/economic insight (e.g., industry averages for insurance, trends like fuel cost).
 - Interest in year-over-year cost/rate trends and surfacing best times or options.

Suggested Improvements & Reviewer Insights

- Shipment Type should include speed (overnight, 2-day) and allow for multiple shipments.
- A deadline-driven input ("when should this arrive?") could auto-suggest shipping options.
- Clarity requested for "protected" terminology and rating meanings.
- Calls for better interaction cues and more intuitive breakdowns (e.g., percent above/below market, direct comparisons).
- Show more direct, card-based price fairness analysis without user needing to click.

Visual/UI Feedback

- Reviewer preferred:

- 1st option: Layout of shipping detail boxes, less daunting to fill out, though wanted more interaction cues (e.g., draggable circles for sliders).
- 2nd option: Clearer, larger shipper logos, better shipper name prominence, more instructive price fairness analysis.
- Color use: Discussion if too much, but largely positive if consistent and helpful as cues.
- Final preference: Combine layout from option 3 with shipper display from option 2.

3. Freight Negotiation Page

Functionality Overview

- For carriers quoting/negotiating shipment rates.
- Inputs: Origin, Destination, Total Distance, Deadhead (empty miles), Offer Rate, Rate per Mile, Fuel Price, Vehicle Type, Weight, Urgency.
- Automated Fairness Analysis showing if pay is in line with market.
- AI-generated negotiation counteroffers and market insights (e.g., rate trends).
- Profit breakdown available (gross, costs, margins).
- Automatic chat/AI assistant for support.
- Additional proposed features: Map view of route.

Reviewer Questions & Feedback

- Clarity requested on "urgency" and standard ranges ("how does this affect my options?").
- Reviewer prefers numerically defined terms (e.g., "urgent = 1-2 days").
- For UI options:
 - Better experience with simple, clear layouts (second layout), particularly for data entry — less intimidating.
 - Color-coded indicators (e.g., below market = red) are effective.

- For star/confidence ratings—preferred first design’s clarity in separating reviews from negotiation acceptance likelihood.
- Disliked four-column layout in th

ird option for being overwhelming.

Overall UI Preferences

- Second option: Inviting data input fields, easy flow.
- Strong preference for color cues, clear interactions, and actionable AI suggestions.
- Wishes for highlights (e.g., green for good margins) to quickly surface performance.

4. Community Dashboard

Main Features

- Directory of shippers, carriers, brokers, verified with community input and external data (e.g., Google Reviews).
- Distinguishes company types (carrier, broker, shipper).
- Cards show key stats (volume, rates, safety).
- Search/filter by company or geolocation proposed.
- Tabulation—pagination at bottom, stats aggregation at top.

Reviewer Perspective

- Initial confusion about dashboard purpose, clarity on company types needed.
- Emphasis on making card interactions obvious (clickable for profile navigation).
- Request for learning tools/help icons to explain terminology and features, aiding new users.
- Importance of search and local filters (e.g., “companies near Pittsburgh”).
- Preference for stat uniformity across same company types.

- Liked tabulated, structured designs (first option felt best, with clear grid and visual order).
- Cards should display same kinds of information for comparability.

5. Company Profile Page

Highlighted Information

- Company summary at top.
- Keywords or "trending" comments surfaced (e.g., "transparent pricing").
- Ratings breakdown (fairness, communication), ideally matching community feedback metrics.
- Comparative stats vs. market average.
- Community reviews/insights and load more/expansion features.
- Additional economic indicators (fuel price index, others).

Reviewer Feedback

- Confused by the "top keyword" term—suggests "top community comment" or similar for clarity; worries users may think it's a search keyword.
- Calls for consistent labeling and clarity on what "fairness" and rating metrics specifically mean (industry average vs. community rating—s

ould be clear).

- Requests:
 - Info/hover icons for jargon or new concepts.
 - Direct connection from rating categories (e.g., stars) to comment sources.
 - Rethink biasing potential of keyword clouds when users write reviews.
 - Use consistent names for similar features (e.g., "community insights" vs. "community reviews").
- Liked community comparison (market average next to rating).

- Liked use of color and friendlier appearance in the second page layout.
- Inquired about the relevance of economic stats (fuel index).
- Cautioned moderation may be needed for user-generated reviews/comments (competitor manipulation risk).

Additional Themes & General Feedback

- **Consistency:** Crucial for labels, metrics, and terminology across all pages.
- **User Education:** Learning curve needs to be addressed — tooltips, explanations, and overlays recommended.
- **Interaction Cues:** Clear indications for clickable elements, interactive sliders, collapsible sections, etc.
- **Accessibility:** Color choices should enhance clarity but not cause confusion.
- **Scalability & Filtering:** Robust filtering/search, especially by company type and location, will enhance usefulness and usability.

Action Items

1. **Redesign Interactive Elements:**
 - Add visual cues showing interactivity (e.g., draggable circles, clickable cards) on all relevant components.
 - Merge preferred visual treatments as described (e.g., shipping details from one version, shippers from another).
2. **Improve Comparisons:**
 - Show price fairness/market comparison data clearly for each shipper with direct, at-a-glance metrics (e.g., percentage above/below market average, as part of each card).
3. **Clarify Terminology:**
 - Standardize terms for cost/rating/fairness metrics.
 - Revise or reword ambiguous labels (“top keyword,” “community insights,” etc.).
4. **Add User Guidance:**

- Incorporate info/help icons and tooltips to explain industry-specific terms or new features
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- Consider an onboarding walkthrough for new users.
1. **Uniformity in Card Design:**
 - Ensure company type cards display consistent core metrics for easier comparison.
 - Make card click targets clearly visible.

Extracted Interview Questions

Shipment Details and Rate Comparison

1. “What does this first section of the app mean to you?”
2. “When we look at the origin zip code, what would you think to input there?”
3. “And then for destination zip code — the final destination?”
4. “The shipment type — what would you think this dropdown should provide you?”
5. “If you look at this tab, the weight tab — what would this be for you?”
6. “How would you expect this bar (package importance) to function?”
7. “What does package importance represent to you?”
8. “What do you think is missing in this section?”
9. “Would shipment type include parcels or something else?”
10. “Would speed type be something we should add?”
11. “What do you think about how package importance connects to rates?”
12. “Would you want to see price ranges tied to that slider?”
13. “How do you interpret insurance or ‘protected’ shipment indicators?”
14. “Would you want to know if delivery time is average or guaranteed?”
15. “Anything worthwhile adding to the rate comparison section?”
16. “What does ‘Explain My Rate’ mean to you?”
17. “What does ‘Show Cost Breakdown by Variable’ mean to you?”

18. "Would that differ from 'Explain My Rate'?"
 19. "How would you interpret 'Price Fairness Analysis'?"
 20. "What level of visual detail (bars, dots) would you prefer for fairness?"
 21. "Which layout do you like more between the three options?"
 22. "Why do you prefer that one?"
 23. "Do you find the screen too busy or detailed?"
 24. "Would limiting clicks improve the experience?"
 25. "How do you interpret the trade-off between clean design vs. information density?"
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Freight Community Profile Page

1. "When you look at the top, what do you think or expect when you see the metrics?"
 2. "What does the term 'verified' mean to you?"
 3. "How do you interpret 'fairness,' 'reliability,' and 'reviews'?"
 4. "Would 'price fairness' make sense as a term?"
 5. "Do you think an info icon or tooltip explaining fairness would help?"
 6. "What does 'Total Insights' mean to you?"
 7. "Would you expect it to match total reviews?"
 8. "How would you interpret the fairness trend?"
 9. "Do these metrics make sense to you as a shipper?"
 10. "Would a toggle help focus on what data you want to see?"
 11. "Do you know what 'broker average' means?"
 12. "Should metrics differ depending on the user type (e.g., shipper vs carrier)?"
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Share Insights Section

1. "What are your thoughts when you look at this section?"

2. "Do the role options (shipper or carrier) make sense?"
 3. "Is the wording 'I participated in this shipment' clear?"
 4. "Would you prefer clearer labels like 'I was the carrier'?"
 5. "Do the fields (route, origin, destination, service date, etc.) feel appropriate?"
 6. "Is 100 characters enough for a comment?"
 7. "How do you interpret 'trending keywords'?"
 8. "What are your thoughts on community insights below?"
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Market and Peer Comparison

1. "What do you think of the market and peer comparison section?"
 2. "Would renewable or sustainable energy usage metrics be meaningful?"
 3. "Would fuel type or carbon footprint matter in explaining fairness differences?"
 4. "Anything else you'd add or adjust for this page?"
 5. "Would this be more effective as a mobile or desktop app?"
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Freight Community Page (Options 1–3)

1. "Which profile layout (1–3) do you prefer?"
 2. "Why do you prefer that one?"
 3. "Would you like interactive buttons for each trend?"
 4. "Do you like the A–B rating system?"
 5. "Would you rather see one performance overview or multiple detailed graphs?"
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Freight Community Dashboard

1. "Is the company directory intuitive?"
 2. "Do you like how metrics differ (volume per month, fleet size, etc.)?"
 3. "Would you prefer standardized metrics?"
 4. "What's your view on highlighting things like 'premium' or 'verified'?"
 5. "Which dashboard option do you prefer and why?"
 6. "Do payment metrics (average payment days) matter to you?"
 7. "Would you rather see operational metrics (volume, reliability) instead?"
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Freight Negotiation Page

1. "What do you think of the Freight Negotiation Assistant?"
2. "Does the AI counteroffer suggestion make sense?"
3. "Does this page seem more tailored to drivers or shippers?"
4. "Would you find value in seeing modest vs aggressive counter suggestions?"
5. "Would this feature help contractors or carriers better negotiate?"

6 - Second Iteration

Figma Link:

<https://www.figma.com/design/aQGu7Hnc8LPVmYHQikwXei/3rd-Party-Trucking-Pricing-Transparency-Software?node-id=0-1&t=3L1nIgPDSLgtoBih-1>